Model Fit

Money Ballers

2025-11-29

```r
pacman::p_load(arrow, lme4, lmerTest, performance, broom, broom.mixed, patchwork,
               arm, viridis, knitr, jtools, ggeffects, ggdist, gt, tidyverse)

options(scipen = 999)

# Set theme for plots
theme_set(theme_minimal())
```

```r
# Setting repository path
#repo_path <- "/Users/namomac/Desktop/Moneyball_FC/inputs/"
repo_path <- "D:/repos/Moneyball_FC/inputs/"

# Reading data
dat <- read_parquet(str_c(repo_path, "player_df.parquet"))
#dat <- read_parquet("player_df.parquet")
```

Time Centering

- **Year**: Centering at the start of the COVID-19 pandemic anchors the model to the most significant economic shock of the decade. This makes the intercept (β_0) interpretable as the expected market value during the 2020 season, rather than an arbitrary start date (2013). Mathematically, placing "0" near the center of the time series reduces collinearity between the linear, quadratic, and cubic time terms, improving model stability.
  - Q: How does market value change annually relative to the 2020 pandemic baseline?

Grand Mean Centering

- **Age**: Market value is determined by biological career stage (e.g., "is Messi 29 and in his prime?" vs "is Messi 19 and a prospect?"). It does *not* matter if you are the oldest guy on a young team (relative age); the market still sees you as 29. We include the linear term (`age_c`) to catch the general trend and the squared term (`age_sq`) to capture the peak and decline curve.

  - Q: How does market value change as a player moves through their biological career arc (from prospect to prime to veteran)?

- **Foreign Player Proportion**: Since no club has 0% foreign players (lowest is 28%), we shift the zero point to the population average (Grand Mean) of internationalization. The intercept therefore represents a club with 'typical' diversity for the league.

  - Q: Does having a *higher proportion of foreign players than the average club* increase market value?

- **Height**: Height is a static physical trait. The market values a 195cm defender because he is objectively tall, not because he is taller than his teammates. Grand mean centering allows us to interpret the coefficient as "the premium for being 1cm taller than the average soccer player. Keeps the intercept interpretable as the value of an"average height" player.

  - Q: Does being taller than the average player in the league increase market value?

Year centering

- **UEFA** Coefficient = UEFA coefficients suffer from "grade inflation" and formula changes over decades; a raw score of 15.0 in 2013 is not equivalent to 15.0 in 2024. Centering by year isolates relative prestige—how dominant the club was *that specific season* compared to its peers. This removes the confounding effect of temporal trends in the coefficient system itself.

  - Q: Does playing for a club that is currently outperforming the European average boost a player's value?

Year-Club mean centering (relative to teammates)

- **minutes_played_c**: Market value is driven by a player's status within their team. This metric isolates whether a player is a "Starter" (positive value) or a "Bench Player" (negative value) relative to their specific squad in that specific season. It removes the noise of "teams that rotate more" vs "teams that don't," focusing purely on the player's hierarchy in the squad.

  - Q: Does being a key player (playing more than the average teammate) increase market value?

Position centering:

- **Goals, Assists:** A goal scored by a striker is expected; a goal scored by a defender is rare and highly valuable. If we leave `goals` and `assists` as raw counts, the model treats 1 goal the same for everyone, potentially undervaluing defensive contributions. We will group by position as performance expectations vary entirely by role. When we center within the position group, we transform the variable from a raw count into a "performance above expectation" metric. This interprets as the effect of scoring above position expectations.
  - Q: Does scoring more goals than the average player in their position increase market value?" (This distinguishes a goal-scoring defender from an underperforming striker).

Non-centered fixed effects:

Performance Metrics

- **Red and Yellow card:** These variables have a meaningful 0 interpretation, and since our outcome is log transformed, these uncentered variables allow us to make percentage interpretations (i.e., one additional red card is associated with beta% change in market value).
  - Q: What is the direct market penalty for receiving one additional red/yellow card?
- **Position**: Reference group will be attackers.
  - Q: How much less is a defender valued compared to an attacker, holding performance constant?
- **Region**: Reference group European & Central Asia.
  - Q: What is the market premium for players in the 'Europe' region compared to South America?
- **Citizen**: Reference group is non-citizen of the country where the club is registered. The coefficient represents the simple difference in means between the two groups. A coefficient of +0.10 means "Citizens are valued 10% higher than Non-Citizens, all else held equal."
  - Q: Is there a homegrown premium' (or discount) for players who hold citizenship in the club's country compared to foreign players?

Random Intercept and slope

- Players: Yearly market values are nested within players.

  - Q: After controlling for goals and age, how much variance in value is simply due to who the player is (marketing value/reputation)?

- Club: Players are cross-classified within clubs.

  - Q: How much of the variation in value is due to the baseline prestige of the club they play for?

- Year: Without this random slope, the model assumes all clubs grow at the same rate. By including it, we allow for "Incubator Clubs" (steep growth) and "Stagnant Clubs" (flat/declining growth).

  - Q: Does the rate of growth in player market value differ significantly between clubs?

Why cross-classified model?

Players can change teams over their span of their careers meaning that their player context can change. They may be a star player in one team and a bench player on a different team that they switch to. In a cross-classified model, the player is the constant unit (i), but the club (j) changes over time (t). If player (I) plays for club (j1) in year (t0) but switches to club (j2) in year (t1) the fixed effect in the model, such as minutes played, updates the players status, as opposed to having static or average predictors. Club attributes, such as foreign players and UEFA scores, ensures that when player move clubs, the model swaps the values of X__j1, for X__j2. Player's characteristics, for instance age and height, move with the player across time.

Assumptions about the data:

We log transform market value to 2024 USD because financial data follows a pareto distribution rather than a normal distribution. If we keep the outcome on the raw scale, our residuals will be extremely large for high value players, and small for low value players violating the assumption of homoscedasticity. Log transforming the outcome compresses the extreme outliers by pulling the long right tail to try and make the distribution normal. We handle exchange rate fluctuation by using historical exchange rates to convert our values to USD.We also adjust market value to 2024 USD because 10 million in 2013 is not the same as in 2024 due to inflation, and by keeping the raw values we would lose real economic value. Therefore, this adjustment allows us to compare real economic power of a player in 2013 versus 2014 without the noise of currency fluctuation.

We assume that variables such as height and positions are time invariant for the duration of this study. At the club level, we assume mutually exclusiveness in that players can only belong to one club in a given year. Although a player could have switched clubs mid-season, we keep this constant for model simplicity. We assume that the relationship between age and log-market value is non-linear and parabolic so we introduce a quadratic term on age. Because we excluded observations with missing data (listwise deletion), we assume the data are Missing Completely at Random (MCAR) or that the proportion of missing cases (~2%) is negligible and does not introduce systematic bias.

Table 1: Data Structure for fixed effects, except year as it is a random slope

Variable Type	**Variable**	Centering Strategy/ Form Factor	Justification
Time	Season Year	Scale-Centered (+ Quadratic)	Centered at 2020, pandemic
Contextual, Club	UEFA Coefficient	Year-Mean Centering	Isolates relative prestige; removes historical inflation
	Foreign Players %	Grand-Mean Centering	Moves baseline to "Average Club" (since 0% is unlikely)
Player Characteristics	Age	Grand-Mean (+ Quadratic)	Captures absolute career stage (Prospect → Prime → Veteran)
	Height	Grand-Mean	Captures premium for absolute physical size
	Position	Reference group, attacker	Attackers tend to be valued more on average
Player Demographic	Citizen of Club Country	None (Raw binary 0/1)	Keeps "Non-Citizen" as clear reference group. Separates individual status from club composition (`foreign_players`)
	Region of origin	Reference group, Europe	Control variable for where player is from
Player Performance	Goals	Position-Centering	Preserves natural 0 and marginal value of 1 goal interpretation
	Assists	Position-Centering	Preserves natural 0 and marginal value of 1 assist interpretation
	Red Card	raw	Meaningful 0 interpretation
	Yellow Card	raw	Meaningful 0 interpretation

Variable Type	Variable	Centering Strategy/ Form Factor	Justification
	Average minutes played in season	Year-Mean Centering, z-score	Isolates whether player is a starter or bench player

Figure 1: Cross-Classified Data Structure: Players move between clubs over time.

```
player_df <- dat |>
  mutate(
    player_id = factor(player_id),
    club_id   = factor(club_id),
    # reference groups
    position = fct_relevel(position, "Attack"),
```

```r
    Region = fct_relevel(Region,  "Europe & Central Asia"),
    # outcome on log scale
    usd_revenue_2024_log = log(usd_revenue_2024),
    # season year
    # scale years with center 2020 for pandomic shock
    season_year_c = scale(season_year, center = 2020),
    # grand mean centering
    foreign_players = foreign_players - mean(foreign_players, na.rm=TRUE),
    age_c = Age - mean(Age, na.rm=TRUE),
    height_c = height_in_cm - mean(height_in_cm, na.rm=TRUE)
  ) |>
  # year mean centering
  group_by(season_year_c) |>
  mutate(uefa_coefficient = uefa_coefficient - mean(uefa_coefficient, na.rm=TRUE),
         minutes_played = scale(minutes_played)) |>
  # position centering
  group_by(position) |>
  mutate(goals = goals - mean(goals, na.rm=TRUE),
         assists = assists - mean(assists, na.rm=TRUE)) |>
  ungroup()
```

- After checking missing values, we saw no more than 108 missing cases for Region, 103 for citizen of club country, and we decide to drop these cases as there are few and we are left with 7,023 cases.

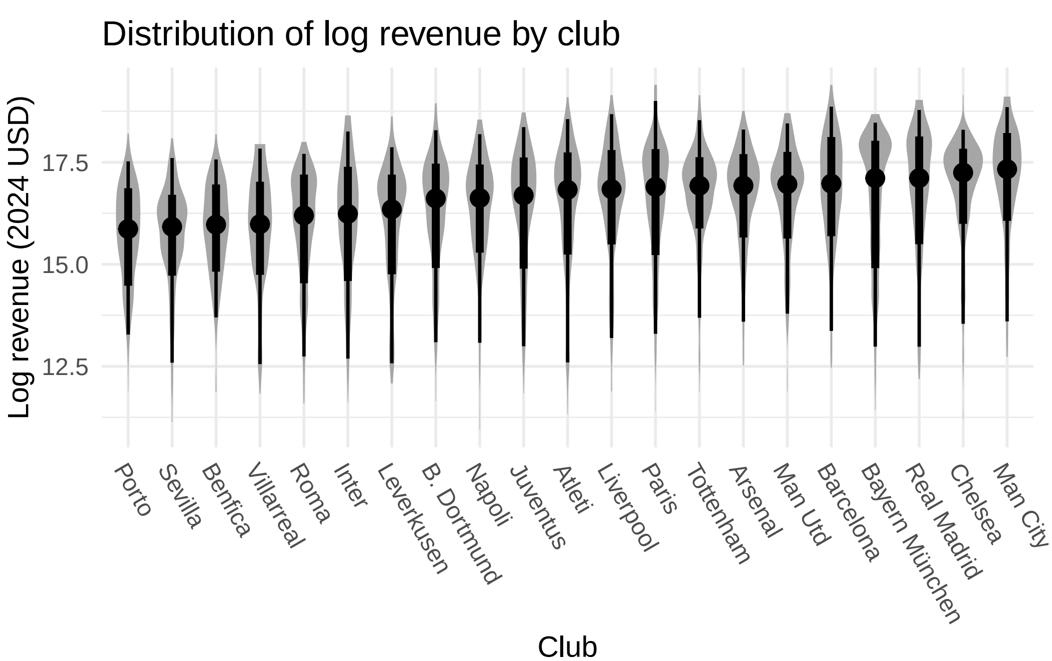
```r
# Which variable has missing value
player_df |>
  reframe(across(everything(), ~sum(is.na(.)))) |>
  select_if(~ . > 0) |>
  pivot_longer(everything(), names_to = "Variable", values_to = "Missing Cases") |>
  slice(1:5) |>
  arrange(desc(`Missing Cases`)) |>
  kable()
```

Variable	Missing Cases
country_of_birth	108
country_of_citizenship	103
height_in_cm	36
date_of_birth	2
Age	2

```
clean_player_df <- player_df |> drop_na()
```

- After examining the distribution of log market values across clubs, it is evident that the boxplots differ in both central tendency and variability. This indicates that club is a meaningful factor in explaining variation in market values.

```
clean_player_df |>
  mutate(Club = fct_reorder(factor(Club), usd_revenue_2024_log, median)) |>
  ggplot(aes(x=Club, y=usd_revenue_2024_log, group=Club)) +
  stat_slabinterval(side = "both")  +
  theme(axis.text.x=element_text(angle = -60, hjust = 0)) +
  labs(title = "Distribution of log revenue by club",
       y="Log revenue (2024 USD)")
```



Linearity

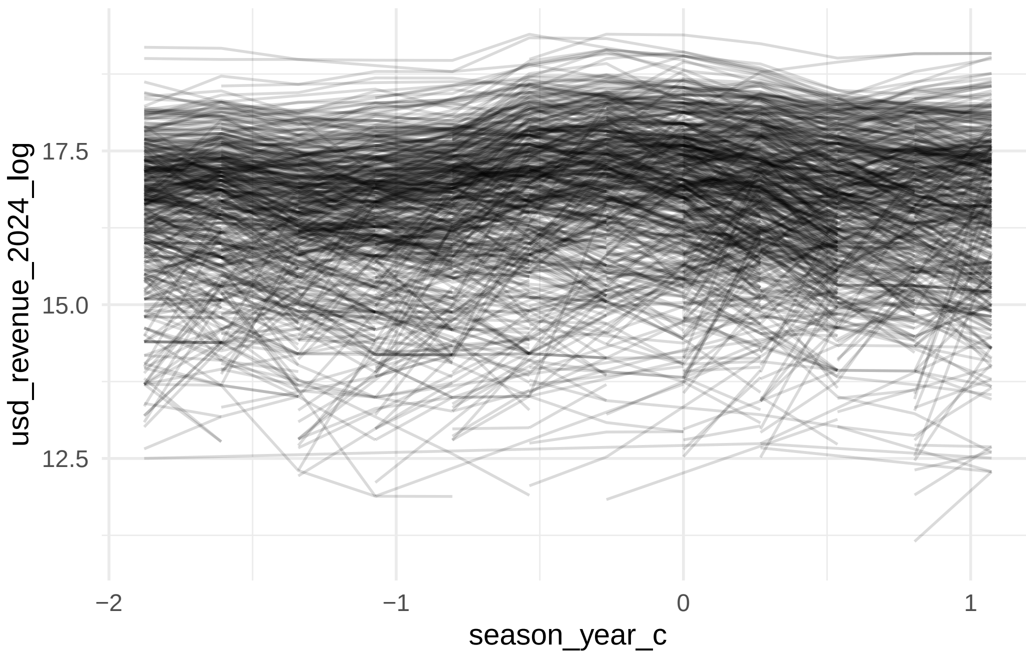
In the overall spaghetti plot group by play_id, the lines vary widely across players, showing considerable between-player heterogeneity in log market value. Within most players, the slopes appear mostly flat, but some of them change up and down dramatically.

In the overall spaghetti plot group by club_id, they are consistently show differences.This means between-club differences in market values are large, supporting the idea that club-level random intercepts are necessary in your multilevel model.

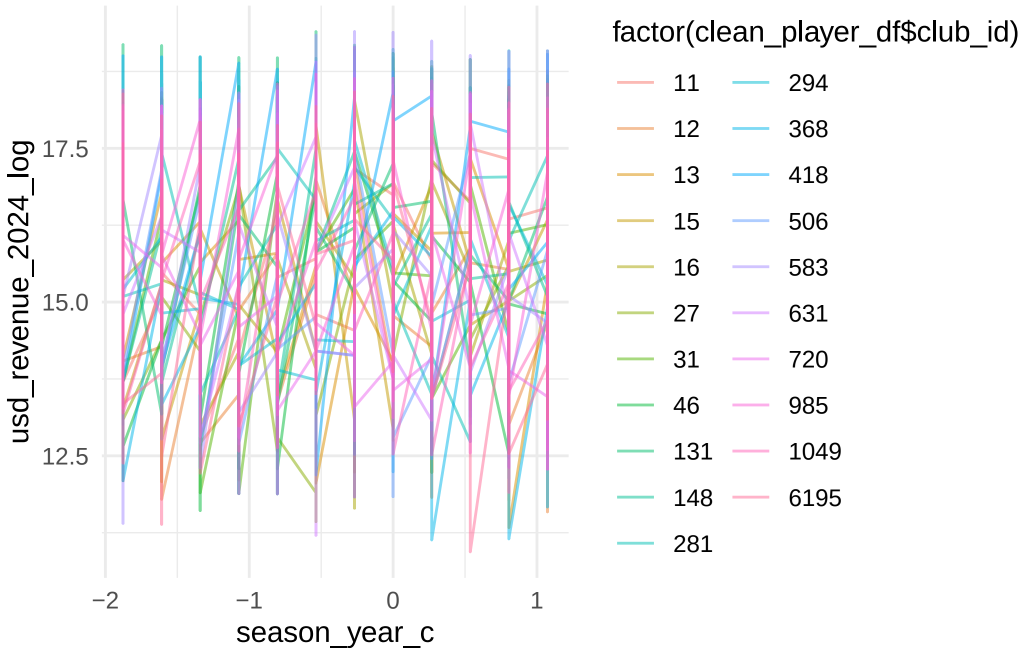
Besides,the clubs' average values change over time, indicating slopes might vary across clubs.

The loess smoother plot shows a non-linear averaged trend (curvature) in log-transformed market value (usd_revenue_2024_log) over centered season year (season_year_c). We can clearly see a peak at about season year 6 to 7, and the shape of the loess line is a wave pattern. Thus, we can try to add polynominal terms in our model to better explain the data.

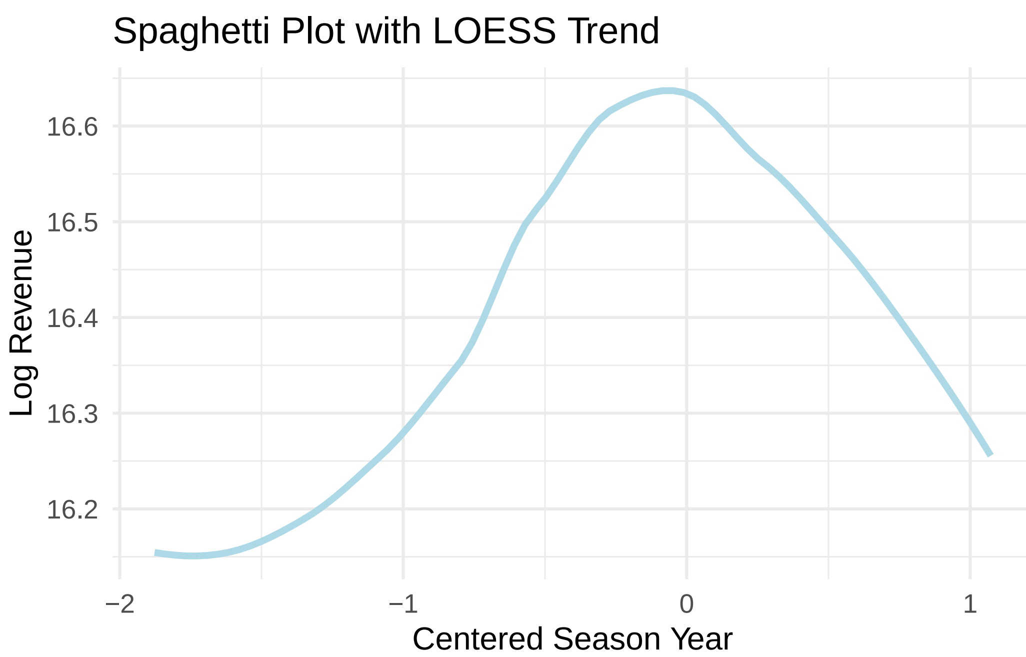
```
ggplot(clean_player_df) +
  geom_line(aes(x = season_year_c, y = usd_revenue_2024_log,
            group = player_id),color = "black", alpha = 0.15)
```




```
ggplot(clean_player_df) +
  geom_line(aes(x = season_year_c, y = usd_revenue_2024_log,
            group = club_id,color = factor(clean_player_df$club_id)),
            alpha = 0.5)
```




```r
ggplot(clean_player_df, aes(x = season_year_c,
                            y = usd_revenue_2024_log)) +

  # LOESS smoother (overall trend)
  geom_smooth(method = "loess",
              color = "lightblue", size = 1.2, se = FALSE) +
  labs(x = "Centered Season Year",
       y = "Log Revenue",
       title = "Spaghetti Plot with LOESS Trend") +
  theme_minimal(base_size = 12)
```



Modeling Process Plan

- Model 1: Random Intercepts only to examine ICC
- Testing whether a random slope on year is needed.
  - Model 2.1: Random intercepts + fixed effect of centered season_year
  - Model 2.2: Adding quadratic terms for season year
  - Model 2.3: Adding cubic term for season year
  - Model 2.4 Adding random slope of season_year_c
  - Model 2.5: Adding quadratic term of season_year as a random slope
- Adding level fixed effects
  - Establish whether age quadratic term is needed
    * Model 3.1: Adding player covariates {Height + Position + Citizen Status + Region + Age + Goals + Assists + Red Cards + Yellow cards + Minutes played}

        * Model 3.2: Adding the quadratic term for age
    - Model 4: Adding club covariates {UEFA coefficients + Foreign Players}

- Compare model fit
- Model diagnostic
- Choose final model

Model 1, intercepts only

```
#Using bobyqa to handle the complex random effects structure and avoid gradient warnings
strict_control <- lmerControl(optimizer = "bobyqa",
                              optCtrl = list(maxfun = 200000))

Model_1 <-  lmer(
  usd_revenue_2024_log ~
    # random effects
      (1|player_id) + (1|club_id), ,
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_1)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20105.1893
BIC	20132.6171
Pseudo-R^2 (fixed effects)	0.0000
Pseudo-R^2 (total)	0.7799

- We can see that player_id accounts for about 75% of market value while club_id accounts for 3%.

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.8205	0.0691	228.8952	26.9891	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3209
club_id	(Intercept)	0.2837
Residual		0.7176

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7455
club_id	21	0.0344

Model 2.1: Adding fixed effect of year

```
Model_2.1 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c +
    # random effects
      (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_2.1)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20050.1704
BIC	20084.4551
Pseudo-R^2 (fixed effects)	0.0048
Pseudo-R^2 (total)	0.7907

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.7735	0.0692	227.8957	27.7128	0.0000
season_year_c	-0.1158	0.0143	-8.1131	6524.0329	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3441
club_id	(Intercept)	0.2822
Residual		0.7088

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7563
club_id	21	0.0333

Model 2.2 Adding quadratic term

- Notice that the linear relationship of year is no longer significant.

```
Model_2.2 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 2) +
    # random effects
      (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_2.2)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20022.0497
BIC	20063.1914
Pseudo-R² (fixed effects)	0.0058
Pseudo-R² (total)	0.7913

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.8250	0.0689	229.5240	27.4536	0.0000
poly(season_year_c, 2)1	-8.9993	1.1102	-8.1060	6520.0361	0.0000
poly(season_year_c, 2)2	-4.0026	0.8992	-4.4510	5650.5048	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3434
club_id	(Intercept)	0.2818
Residual		0.7076

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7568
club_id	21	0.0333

Compare Model 2.1 and 2.2 to determine if the quadratic term is needed.

- We see that the LRT test suggests that the full model provides a significantly better fit than the reduce model, the quadratic term is needed for this data.

```r
anova(Model_2.1, Model_2.2, test = "LRT") |> gt()
```

npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
5	20039.98	20074.27	-10014.99	20029.98	NA	NA	NA
6	20022.19	20063.33	-10005.10	20010.19	19.78947	1	0.000008645778

Model 2.3 Adding cubic term

- Now linear year trends is significant

```
Model_2.3 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3) +
    # random effects
      (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_2.3)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19996.7231
BIC	20044.7217
Pseudo-R^2 (fixed effects)	0.0067
Pseudo-R^2 (total)	0.7930

Compare Model 2.2 and 2.3 to determine if the quadratic term is needed.

- We see that the LRT test suggests that the full model provides a significantly better fit than the reduce model, the cubic term is needed for this data. After adding cubic terms,

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.8258	0.0690	229.3910	27.4814	0.0000
poly(season_year_c, 3)1	-8.9114	1.1079	-8.0432	6502.3364	0.0000
poly(season_year_c, 3)2	-4.0680	0.8969	-4.5354	5640.4407	0.0000
poly(season_year_c, 3)3	-4.2197	0.8282	-5.0949	5343.9717	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3458
club_id	(Intercept)	0.2819
Residual		0.7055

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7583
club_id	21	0.0333

it yields a better fit. It cannot only explain the peak of the overall trajectory, but can also explain the up and down wave pattern over time of the trajectory.

```
anova(Model_2.2, Model_2.3, test = "LRT") |> gt()
```

Model 2.4, Random slope

- To estimate a random slope `(1 + year_centered | player_id)`, the model needs multiple data points for *every* player to calculate their individual trajectory. But many players only appear in our data for 1 or 2 years, and the model struggles to distinguish the "player slope" from the "residual error," causing these convergence issues. Clubs, however, have many players and many years of historical market value data so a random slope for clubs is more robust to fit.

npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
6	20022.19	20063.33	-10005.096	20010.19	NA	NA	NA
7	19998.32	20046.32	-9992.159	19984.32	25.87479	1	0.0000003642961

```
Model_2.4 <-  lmer(
  usd_revenue_2024_log ~
    ## first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3) +
    # random effects
      (1 | player_id) + (poly(season_year_c, 1) | club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_2.4)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19934.0370
BIC	19995.7495

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.8263	0.0679	233.0169	27.8934	0.0000
poly(season_year_c, 3)1	-8.3753	2.7303	-3.0675	20.0972	0.0061
poly(season_year_c, 3)2	-4.2265	0.8897	-4.7506	5621.1039	0.0000
poly(season_year_c, 3)3	-4.3412	0.8206	-5.2901	5311.7316	0.0000

p values calculated using Satterthwaite d.f.

Model 2.5, Random slope + quadratic term

```
Model_2.5 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
```

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3508
club_id	(Intercept)	0.2762
club_id	poly(season_year_c, 1)	11.4348
Residual		0.6975

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7643
club_id	21	0.0319

```
    # Use poly() with raw=TRUE
    poly(season_year_c, 3) +
    # random effects
      (1 | player_id) + (poly(season_year_c, 2) | club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_2.5)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19909.5925
BIC	19991.8759

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.8263	0.0681	232.3099	27.8542	0.0000
poly(season_year_c, 3)1	-8.4042	2.7601	-3.0449	20.1057	0.0064
poly(season_year_c, 3)2	-4.2759	1.7913	-2.3871	19.3076	0.0274
poly(season_year_c, 3)3	-4.2851	0.8169	-5.2455	5295.4842	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3515
club_id	(Intercept)	0.2772
club_id	poly(season_year_c, 2)1	11.5921
club_id	poly(season_year_c, 2)2	7.1255
Residual		0.6934

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7661
club_id	21	0.0322

```
anova(Model_2.4, Model_2.5, test = "LRT") |> gt()
```

Model 2.6, Random slope + quadratic term + cubic

```
Model_2.6 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3) +
    # random effects
      (1 | player_id) + (poly(season_year_c, 3)| club_id),
  data = clean_player_df, REML = TRUE,    control = strict_control)

summ(digits = 4,  Model_2.6)
```

npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
9	19937.35	19999.06	-9959.673	19919.35	NA	NA	NA
12	19914.28	19996.56	-9945.140	19890.28	29.06578	3	0.000002169278

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19907.9512
BIC	20017.6623
Pseudo-R^2 (fixed effects)	0.0062
Pseudo-R^2 (total)	0.8026

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.8267	0.0682	232.0647	27.8580	0.0000
poly(season_year_c, 3)1	-8.3403	2.7654	-3.0160	20.1197	0.0068
poly(season_year_c, 3)2	-4.2126	1.8037	-2.3355	19.2777	0.0305
poly(season_year_c, 3)3	-4.3503	1.0180	-4.2734	30.4146	0.0002

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3526
club_id	(Intercept)	0.2775
club_id	poly(season_year_c, 3)1	11.6228
club_id	poly(season_year_c, 3)2	7.1933
club_id	poly(season_year_c, 3)3	2.7889
Residual		0.6924

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7668
club_id	21	0.0323

```r
anova(Model_2.3, Model_2.5, Model_2.6, test = "LRT") |> gt()
```

- We see that Model 2.6 with the cubic slope failed to converge as we are asking the model to estimate too many parameters for very few clubs (21 in total). Additionally, it is marginally better than the Model 2.5, but with a higher BIC value. We keep Model 2.5

npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
7	19998.32	20046.32	-9992.159	19984.32	NA	NA	NA
12	19914.28	19996.56	-9945.140	19890.28	94.037187	5	0.0000000000000000009519902
16	19912.78	20022.49	-9940.388	19880.78	9.503629	4	0.0496727390456252748673904

for simplicity.

Determine if the random slope on season year is needed

```
# LRT test for 2 random slopes
calculate_lrt_mixture <- function(reduced_model, full_model) {

  # 1. Extract Log-Likelihoods
  ll_reduced = as.numeric(logLik(reduced_model))
  ll_full = as.numeric(logLik(full_model))

  # 2. Calculate LRT Statistic
  lrt_stat = -2 * (ll_reduced - ll_full)

  # 3. Calculate Difference in Parameters (df)
  df_reduced = attr(logLik(reduced_model), "df")
  df_full = attr(logLik(full_model), "df")
  df_diff = df_full - df_reduced

  # 4. Calculate P-Value based on the difference

  if (df_diff == 3) {
    # --- NEW BLOCK FOR QUADRATIC SLOPE ---
    # Going from 2 Random Effects (Int+Lin) -> 3 Random Effects (Int+Lin+Quad)
    # We add 1 variance + 2 covariances = 3 parameters.
    # The mixture is 0.5 * ChiSq(2) + 0.5 * ChiSq(3)
    p_val = 0.5 * pchisq(lrt_stat, df = 2, lower.tail = FALSE) +
             0.5 * pchisq(lrt_stat, df = 3, lower.tail = FALSE)

    msg = "Mixture of ChiSq(2) and ChiSq(3)"

  } else if (df_diff == 2) {
    # Going from 1 Random Effect (Int) -> 2 Random Effects (Int+Lin)
    # Adds 1 variance + 1 covariance = 2 parameters
```

```r
    p_val = 0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE) +
              0.5 * pchisq(lrt_stat, df = 2, lower.tail = FALSE)

    msg = "Mixture of ChiSq(1) and ChiSq(2)"

  } else if (df_diff == 1) {
    # Adding a single parameter (e.g. just a variance, no covariance)
    p_val = 0.5 * pchisq(lrt_stat, df = 0, lower.tail = FALSE) +
              0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE)

    msg = "Mixture of ChiSq(0) and ChiSq(1)"

  } else {
    # Fallback for standard fixed effects comparisons
    p_val = pchisq(lrt_stat, df = df_diff, lower.tail = FALSE)
    msg = paste0("Standard ChiSq(", df_diff, ")")
  }

  # Output
  cat("LRT Statistic:", round(lrt_stat, 4), "\n")
  cat("Parameter Diff:", df_diff, "\n")
  cat("Distribution:  ", msg, "\n")
  cat("P-Value:       ", format.pval(p_val, eps = .0001), "\n")
}

# Run the comparison
calculate_lrt_mixture(Model_2.3, Model_2.5)
```

```
LRT Statistic: 97.1306
Parameter Diff: 5
Distribution:   Standard ChiSq(5)
P-Value:        < 0.0001
```

- To determine if the rate of market value growth varies significantly across clubs, we compared the random-intercept only model with season_year_c linear and quadratic fixed effects to the model with a random linear and quadratic slopes using a Likelihood Ratio Test (LRT). The Null hypothesis tests whether the variance parameter on the boundary of the parameter space ($H_0 : \sigma^2_{slope} = 0$), we utilized a helper function to evaluate significance against a correct 50:50 mixture distribution of $.5 \times \chi^2(1) + .5 \times \chi^2(2)$. The test yielded a significant results ($\chi^2 = 97.13, p < .001$) leading us to reject the Null. This provides strong evidence that clubs do not follow parallel trends; rather, there is significant heterogeneity in club trajectories, justifying the inclusion of the random slope.

Model 3.1: Adding fixed effects at level 1 with age as a linear relationship

- Age centered is not even significant. Same for red\yellow cards.

```
Model_3.1 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3) +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    age_c +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # random effects
    (1 | player_id) + (poly(season_year_c, 2) | club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_3.1)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	17405.6747
BIC	17597.6692

Model 3.2. Adding age quadratic term

```
Model_3.2 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3) +
    # player fixed effects
```

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.4847	0.0831	198.4536	48.8616	0.0000
poly(season_year_c, 3)1	-4.2451	2.4394	-1.7402	24.7065	0.0943
poly(season_year_c, 3)2	-7.1436	1.5091	-4.7336	19.4214	0.0001
poly(season_year_c, 3)3	-4.7445	0.7122	-6.6620	5223.6686	0.0000
height_c	0.0142	0.0038	3.7960	1672.1013	0.0002
positionDefender	-0.4285	0.0596	-7.1860	1652.1403	0.0000
positionGoalkeeper	-1.2393	0.0977	-12.6878	1668.7200	0.0000
positionMidfield	-0.2233	0.0596	-3.7482	1680.6181	0.0002
citizen_of_club_countryYes	-0.3962	0.0375	-10.5595	5744.5210	0.0000
RegionEast Asia & Pacific	-0.1702	0.2112	-0.8058	1735.7054	0.4204
RegionLatin America & Caribbean	0.2493	0.0634	3.9308	1799.0985	0.0001
RegionMiddle East & North Africa	0.0868	0.2453	0.3540	1699.5636	0.7234
RegionNorth America	0.0071	0.2250	0.0314	1779.9129	0.9749
RegionSub-Saharan Africa	0.0219	0.1039	0.2107	1734.3466	0.8331
age_c	-0.0006	0.0040	-0.1441	2353.2437	0.8854
goals	0.0139	0.0026	5.3603	5517.1209	0.0000
assists	0.0127	0.0036	3.4888	5055.4381	0.0005
red_cards	-0.0177	0.0295	-0.6009	4691.8114	0.5479
yellow_cards	0.0062	0.0041	1.5211	5382.5589	0.1283
minutes_played	0.5081	0.0173	29.3265	5585.8774	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.9477
club_id	(Intercept)	0.2976
club_id	poly(season_year_c, 2)1	9.6980
club_id	poly(season_year_c, 2)2	5.9455
Residual		0.6110

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.6604
club_id	21	0.0651

```r
    height_c + position + citizen_of_club_country + Region +
    poly(age_c, 2, raw = TRUE) +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # random effects
    (1 | player_id) + (poly(season_year_c, 2) | club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_3.2)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	14425.2126
BIC	14624.0640

- It seems that we need the quadratic term for age in the model.

```r
anova(Model_3.1, Model_3.2) |> gt()
```

Model 4: Adding club covariates

- The model with club covariates under performs compared to model 3.2, and both club covariates are not significant which we keep model 3.2

```r
Model_4 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3) +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    poly(age_c, 2, raw = TRUE) +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # club fixed effects
    uefa_coefficient + foreign_players +
    # random effects
    (1 | player_id) + (poly(season_year_c, 2) | club_id),
```

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.8885	0.0690	244.6560	76.3737	0.0000
poly(season_year_c, 3)1	-1.4829	1.6267	-0.9116	38.3786	0.3677
poly(season_year_c, 3)2	-8.6043	1.3562	-6.3445	19.3932	0.0000
poly(season_year_c, 3)3	-5.9166	0.5402	-10.9535	5018.2007	0.0000
height_c	0.0102	0.0036	2.8111	1800.3844	0.0050
positionDefender	-0.4032	0.0579	-6.9589	1785.9484	0.0000
positionGoalkeeper	-1.0589	0.0949	-11.1603	1797.2313	0.0000
positionMidfield	-0.1903	0.0578	-3.2921	1802.2507	0.0010
citizen_of_club_countryYes	-0.1863	0.0314	-5.9415	6882.5272	0.0000
RegionEast Asia & Pacific	-0.4258	0.2044	-2.0834	1835.1430	0.0374
RegionLatin America & Caribbean	0.1891	0.0610	3.1019	1953.9044	0.0020
RegionMiddle East & North Africa	-0.0959	0.2378	-0.4031	1814.4431	0.6869
RegionNorth America	-0.0579	0.2172	-0.2666	1855.7128	0.7898
RegionSub-Saharan Africa	0.0650	0.1004	0.6474	1848.6130	0.5175
poly(age_c, 2, raw = TRUE)1	0.0188	0.0037	5.0349	2290.4568	0.0000
poly(age_c, 2, raw = TRUE)2	-0.0242	0.0004	-63.1309	6109.0016	0.0000
goals	0.0046	0.0020	2.3416	5141.9433	0.0192
assists	0.0096	0.0027	3.5013	4857.8122	0.0005
red_cards	0.0097	0.0221	0.4370	4666.9432	0.6622
yellow_cards	-0.0015	0.0031	-0.4931	5060.0640	0.6220
minutes_played	0.3497	0.0133	26.2050	5216.0125	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.9626
club_id	(Intercept)	0.2201
club_id	poly(season_year_c, 2)1	5.3998
club_id	poly(season_year_c, 2)2	5.5939
Residual		0.4548

```
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_4)
```

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7840
club_id	21	0.0410

npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
28	17323.68	17515.67	-8633.839	17267.68	NA	NA	NA
29	14323.32	14522.17	-7132.661	14265.32	3002.357	1	0

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	14442.4172
BIC	14654.9825

```
anova(Model_3.2, Model_4) |> gt()
```

```
export_summs(Model_2.5, Model_3.2,
              digits=3, r.squared=FALSE,
             model.names = c("Model_2.5", "Model_3.2"))
```

```
final_model <- Model_3.2
```

Model Diagnostics

- By comparing the QQ-line and QQ-plot (both residuals and random effects) we can find that both model plots look good on normality, except some outliers. By comparing the

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.8887	0.0691	244.3374	75.2306	0.0000
poly(season_year_c, 3)1	-1.4882	1.6441	-0.9052	37.4140	0.3711
poly(season_year_c, 3)2	-8.4400	1.3437	-6.2813	19.7101	0.0000
poly(season_year_c, 3)3	-5.9420	0.5414	-10.9760	5016.9125	0.0000
height_c	0.0102	0.0036	2.8121	1797.7642	0.0050
positionDefender	-0.4035	0.0579	-6.9671	1783.2207	0.0000
positionGoalkeeper	-1.0589	0.0948	-11.1664	1794.6754	0.0000
positionMidfield	-0.1903	0.0578	-3.2937	1799.5170	0.0010
citizen_of_club_countryYes	-0.1850	0.0314	-5.8911	6888.2586	0.0000
RegionEast Asia & Pacific	-0.4258	0.2043	-2.0844	1832.6128	0.0373
RegionLatin America & Caribbean	0.1881	0.0609	3.0870	1951.7913	0.0021
RegionMiddle East & North Africa	-0.0970	0.2377	-0.4080	1811.8061	0.6833
RegionNorth America	-0.0592	0.2171	-0.2727	1853.2961	0.7851
RegionSub-Saharan Africa	0.0637	0.1004	0.6349	1846.4439	0.5256
poly(age_c, 2, raw = TRUE)1	0.0187	0.0037	5.0195	2289.6846	0.0000
poly(age_c, 2, raw = TRUE)2	-0.0242	0.0004	-63.0836	6117.7579	0.0000
goals	0.0047	0.0020	2.3601	5138.2429	0.0183
assists	0.0096	0.0027	3.5140	4853.8040	0.0004
red_cards	0.0093	0.0221	0.4182	4663.1714	0.6758
yellow_cards	-0.0015	0.0031	-0.4904	5054.9730	0.6239
minutes_played	0.3498	0.0133	26.2037	5211.4339	0.0000
uefa_coefficient	-0.0005	0.0012	-0.3860	4152.6167	0.6995
foreign_players	0.1029	0.1038	0.9914	1353.6306	0.3217

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.9620
club_id	(Intercept)	0.2208
club_id	poly(season_year_c, 2)1	5.5112
club_id	poly(season_year_c, 2)2	5.4796
Residual		0.4550

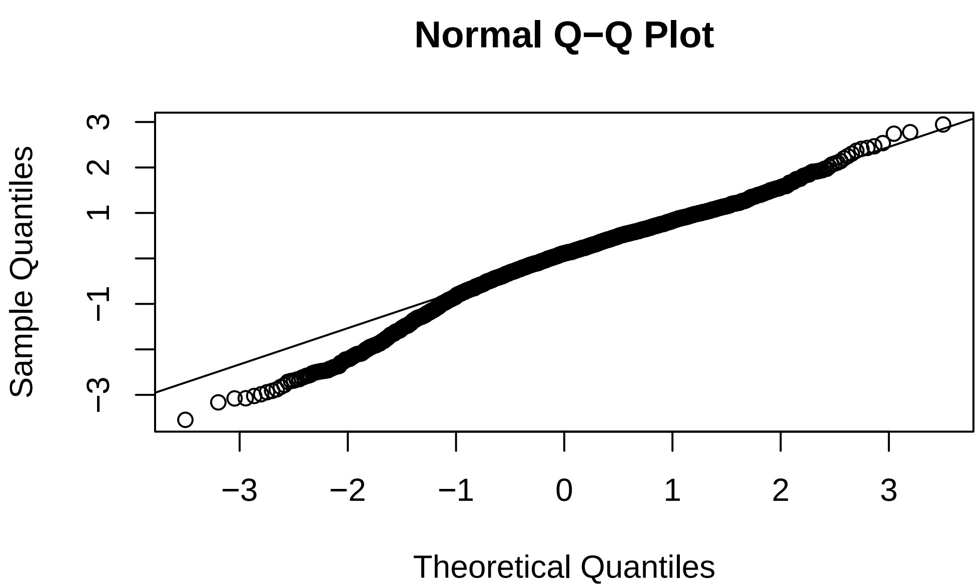
residual-fitted value plot, we can find that both model show the same spread pattern, the dots are evenly spread across zero-mean line up and down.

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7835
club_id	21	0.0413

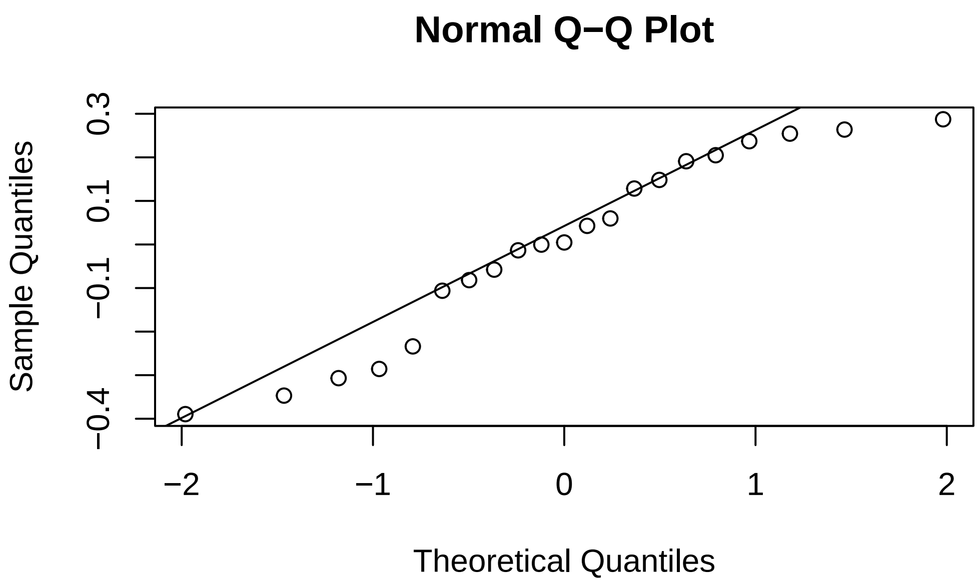
npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
29	14323.32	14522.17	-7132.661	14265.32	NA	NA	NA
31	14326.19	14538.75	-7132.094	14264.19	1.132996	2	0.5675093

- By comparing EBLUPs plots, we can see that most of the points lie deviate from the diagonal at both tails, and the lower side is more downward, suggesting moderate departures from normality in the random slopes and random intercepts, which indicates possible skewness and outliers.Thus, we cannot say the normal random effects assumption in the multilevel model are met.

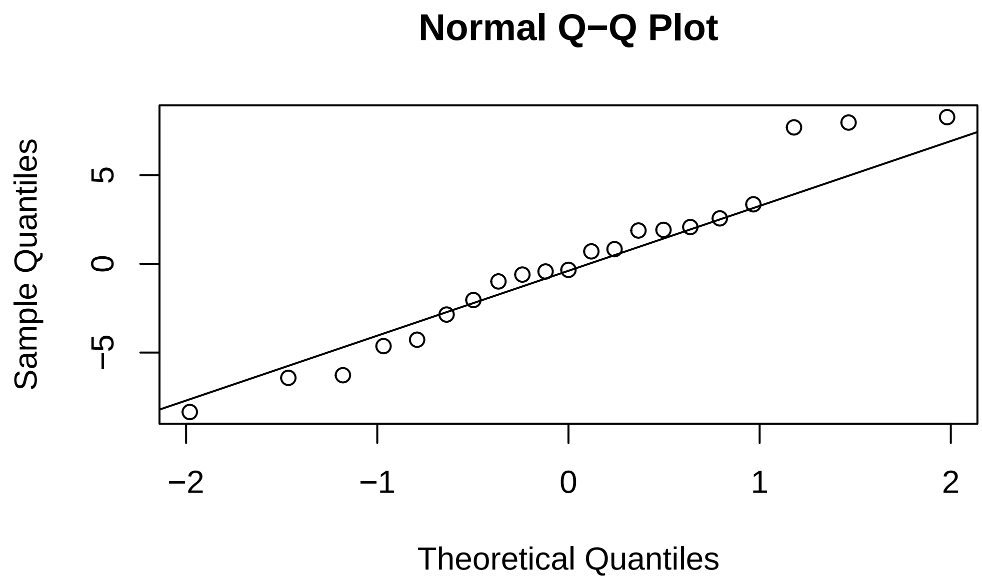
```
# slopes
qqnorm(ranef(final_model)$player_id[,1]) # group factor
qqline(ranef(final_model)$player_id[,1])
```



```r
qqnorm(ranef(final_model)$club_id[,1]) # group factor
qqline(ranef(final_model)$club_id[,1])
```



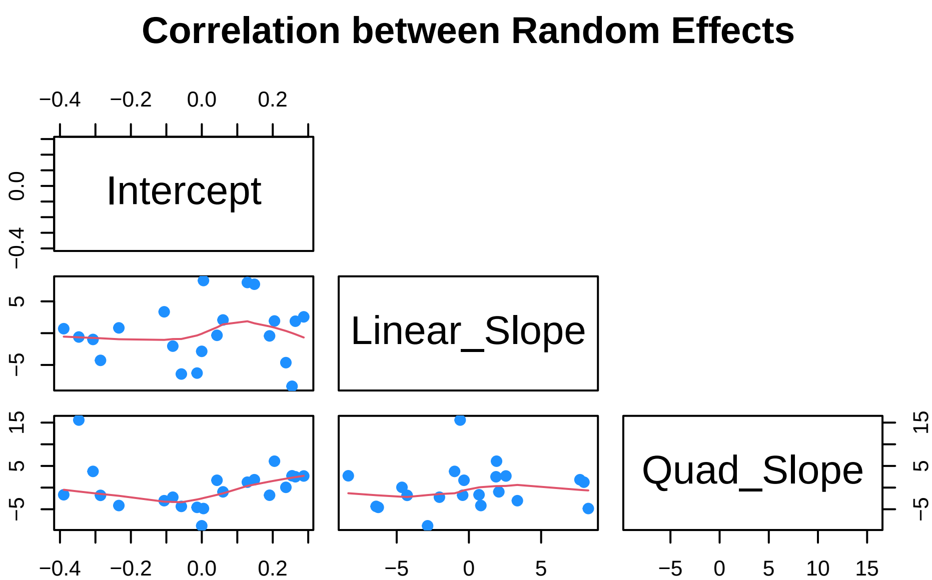

```r
qqnorm(ranef(final_model)$club_id[,2]) # group factor
qqline(ranef(final_model)$club_id[,2])
```



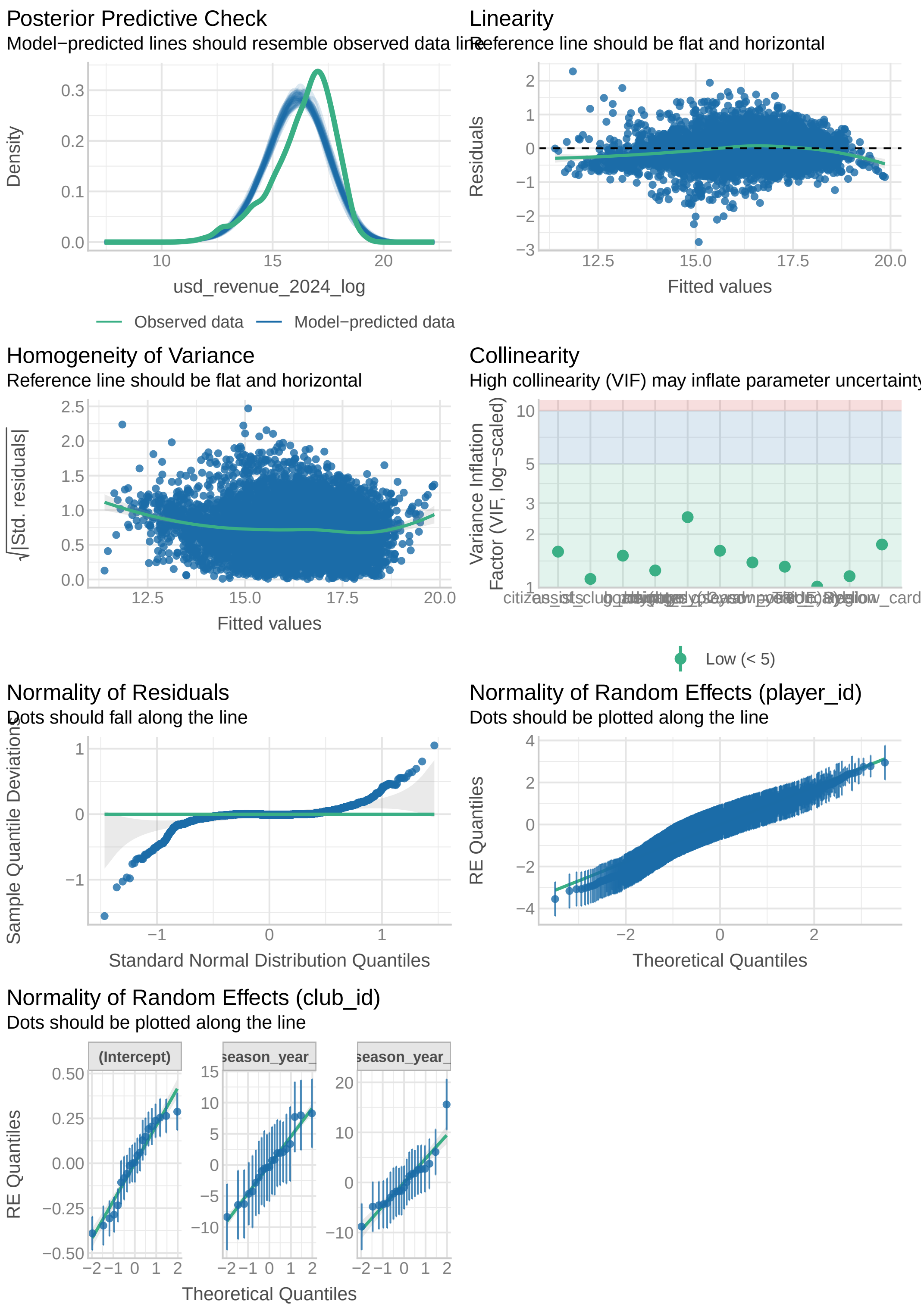
- The red regression line slopes slightly downward, but only gently. That is, clubs with higher baseline log market value tend to have slightly more negative or flatter slopes. Clubs with lower baseline market value tend to have slightly steeper slopes.
- Most residuals lie between about -1 and $+1$, and fitted log revenues between 14 and 18. There is no strong curved pattern or obvious systematic relationship between residual and fitted values. So the residual reasonably consistent with the linearity assumption. Besides, there is no funnel shape, so the residuals are homogeneous. However, we can still see some outliers exist.

```r
# 1. Extract the random effects for clubs
re <- ranef(final_model)$club_id

# 2. Rename columns for clarity (poly names are messy)
# Note: Ensure your model actually has 3 REs. If not, adjust indices.
colnames(re) <- c("Intercept", "Linear_Slope", "Quad_Slope")

# 3. Create a Scatterplot Matrix (Pairs Plot)
# This plots every random effect against every other random effect
pairs(re,
      main = "Correlation between Random Effects",
      pch = 19,
      col = "dodgerblue",
```

```
      lower.panel = panel.smooth, # Adds trend lines automatically
      upper.panel = NULL)         # Hides duplicate upper half
```




```
check_model(final_model)
```


Posterior Predictive Check
Model−predicted lines should resemble observed data line
Density
usd_revenue_2024_log
Observed data
Model−predicted data
Linearity
Reference line should be flat and horizontal
Residuals
Fitted values
Homogeneity of Variance
Reference line should be flat and horizontal
√|Std. residuals|
Fitted values
Collinearity
High collinearity (VIF) may inflate parameter uncertainty
Variance Inflation Factor (VIF, log−scaled)
Low (< 5)
Normality of Residuals
Dots should fall along the line
Sample Quantile Deviations
Standard Normal Distribution Quantiles
Normality of Random Effects (player_id)
Dots should be plotted along the line
RE Quantiles
Theoretical Quantiles
Normality of Random Effects (club_id)
Dots should be plotted along the line
(Intercept)
season_year_
season_year_
RE Quantiles
Theoretical Quantiles

Final Model Specification

We specify a **cross-classified multilevel model** (Model 3.2) to partition the variance in market value across observations (t), players (i), and clubs (j).

Level 1: Observation Model (Time-Varying)

At the observation level, the log-transformed market value is modeled as a function of the player-club intercept (π_{0ij}), orthogonal time trends, biological aging, and performance metrics:

$$
\begin{aligned}
\log(\text{Revenue})_{tij} = \pi_{0ij} & \\
& + \pi_{1j} \cdot \rho_1(\text{Year}_t) + \pi_{2j} \cdot \rho_2(\text{Year}_t) + \pi_3 \cdot \rho_3(\text{Year}_t) \\
& + \pi_4(\text{Age}_{it} - \bar{A}) + \pi_5(\text{Age}_{it} - \bar{A})^2 \\
& + \pi_6(\text{Mins}_{Z,it}) + {}_{perf} + \epsilon_{tij}
\end{aligned}
$$

Note: ρ_k(Year) denotes the k-th order orthogonal polynomial for Season Year. Age is modeled using raw polynomials.

Level 2a: Player Model (Time-Invariant)

The player-club intercept (π_{0ij}) is decomposed into a fixed component based on player characteristics and a random player effect (u_{0i}):

$$
\pi_{0ij} = \beta_{0j} + \gamma_1(\text{Height}_i) + \gamma_2(\text{Citizen}_{ij}) + \sum_p \gamma_p(\text{Pos}) + \sum_r \gamma_r(\text{Reg}) + u_{0i}
$$

Level 2b: Club Model (Contextual Trajectories)

The club baseline (β_{0j}) and the time-varying slope parameters (π_{1j}, π_{2j}) are modeled with random effects to allow for heterogeneity in both growth velocity and trajectory curvature:

$$
\begin{aligned}
\beta_{0j} &= \delta_{00} + v_{0j} && \text{(Baseline Prestige)} \\
\pi_{1j} &= \delta_{10} + v_{1j} && \text{(Linear Trend Deviation)} \\
\pi_{2j} &= \delta_{20} + v_{2j} && \text{(Quadratic Curvature Deviation)} \\
\pi_3 &= \delta_{30} && \text{(Fixed Cubic Trend)}
\end{aligned}
$$

Variance-Covariance Structure

$$
u_{0i} \sim N(0, \sigma_u^2) \quad , \quad \begin{pmatrix} v_{0j} \\ v_{1j} \\ v_{2j} \end{pmatrix} \sim N\left(\mathbf{0}, \begin{pmatrix} \sigma_{Int}^2 & \sigma_{01} & \sigma_{02} \\ \sigma_{01} & \sigma_{Lin}^2 & \sigma_{12} \\ \sigma_{02} & \sigma_{12} & \sigma_{Quad}^2 \end{pmatrix} \right)
$$

Interpretation of Results

Model Fit and Structure The final model (Model 3.2) explains substantial variance in market value ($Pseudo\text{-}R^2_{total} \approx 0.89$). The Intraclass Correlation Coefficient (ICC) for players remains high ($\sigma^2_{player} \approx 0.93$), confirming that individual reputation ("The Superstar Effect") is the dominant driver of valuation.

Temporal Dynamics (Orthogonal Trends) The fixed effects for time utilize orthogonal polynomials to ensure stability. The significant cubic term ($\pi_3 = -5.92, p < .001$) and quadratic term ($\pi_2 = -8.60, p < .001$) confirm the "S-shaped" market evolution: stagnation, rapid inflation ("Boom"), and recent deceleration. Notably, we detected significant variation in these trajectories across clubs; the random quadratic slope ($\sigma^2_{Quad} \approx 31.29$) indicates that clubs differ significantly not just in their baseline value, but in the *shape* of their value growth curves (e.g., volatile vs. stable growth).

Biological Career Arc We observe a robust "Inverted-U" career trajectory. The raw quadratic age term is significant and negative ($\pi_5 = -0.024, p < .001$), predicting that market value accelerates during a player's development, peaks during their prime years, and depreciates significantly as they age, independent of their on-field performance statistics.

Squad Status and Performance Squad status remains a massive driver of valuation. The standardized coefficient for minutes played ($\pi_6 \approx 0.35$) suggests that a one standard deviation increase in playing time is associated with a **42% increase** in market value ($e^{0.35} - 1$).

Contextual Premiums At the individual level, we observe a "Domestic Discount." Homegrown citizens are valued approximately **17% lower** than comparable foreign imports ($\gamma_2 = -0.19$; $e^{-0.19} - 1 \approx -0.17$).

```
[1] 26.08771
```

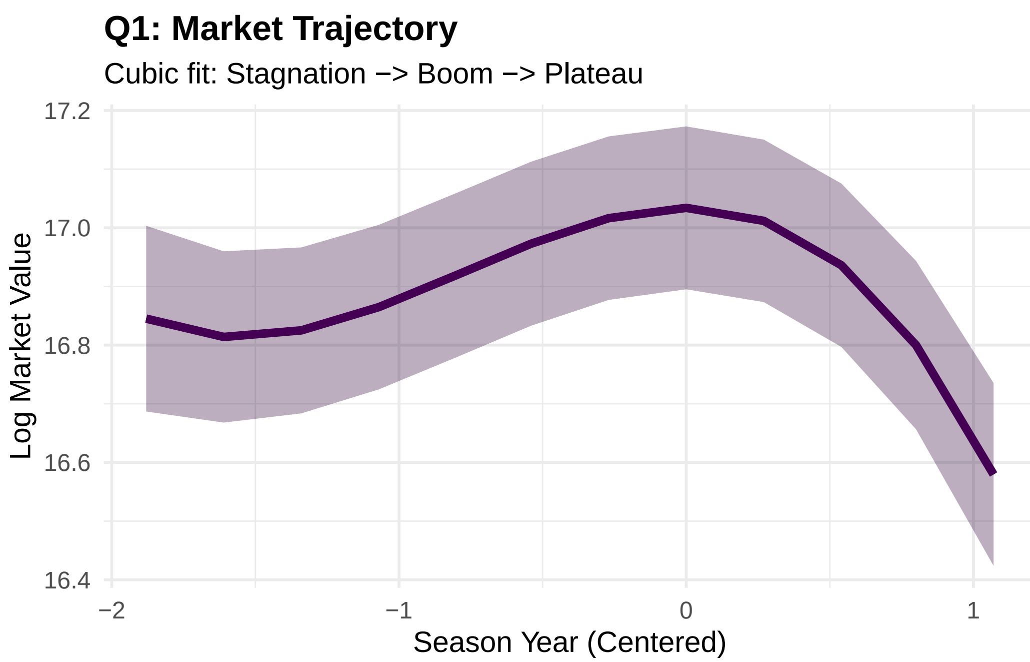
Visualization of Marginal Effects

Q1: Market Evolution

How does market value change annually compared to the 2013 baseline?

- The visualization answers Q1 by displaying a statistically significant S-shaped trajectory ($\beta_{cubic} \approx -0.002$). Rather than constant inflation, the market shows three distinct phases: an initial period of stability (2013-2014), a rapid "Boom" period of hyper-inflation (2015-2019), and a recent "Plateau" or deceleration in the post-2020 era.

```r
# Generate predictions for the full range of years
# [all] tells it to use a dense grid for a smooth curve
pred_trend <- ggpredict(final_model, terms = "season_year_c [all]")

plot(pred_trend) +
  # Custom colors to match your Viridis Purple preference
  geom_line(color = "#440154", linewidth = 1.5) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high),
              fill = "#440154", alpha = 0.2) +
  # Labels and Theme
  labs(
    title = "Q1: Market Trajectory",
    subtitle = "Cubic fit: Stagnation -> Boom -> Plateau",
    x = "Season Year (Centered)",
    y = "Log Market Value"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold"))
```

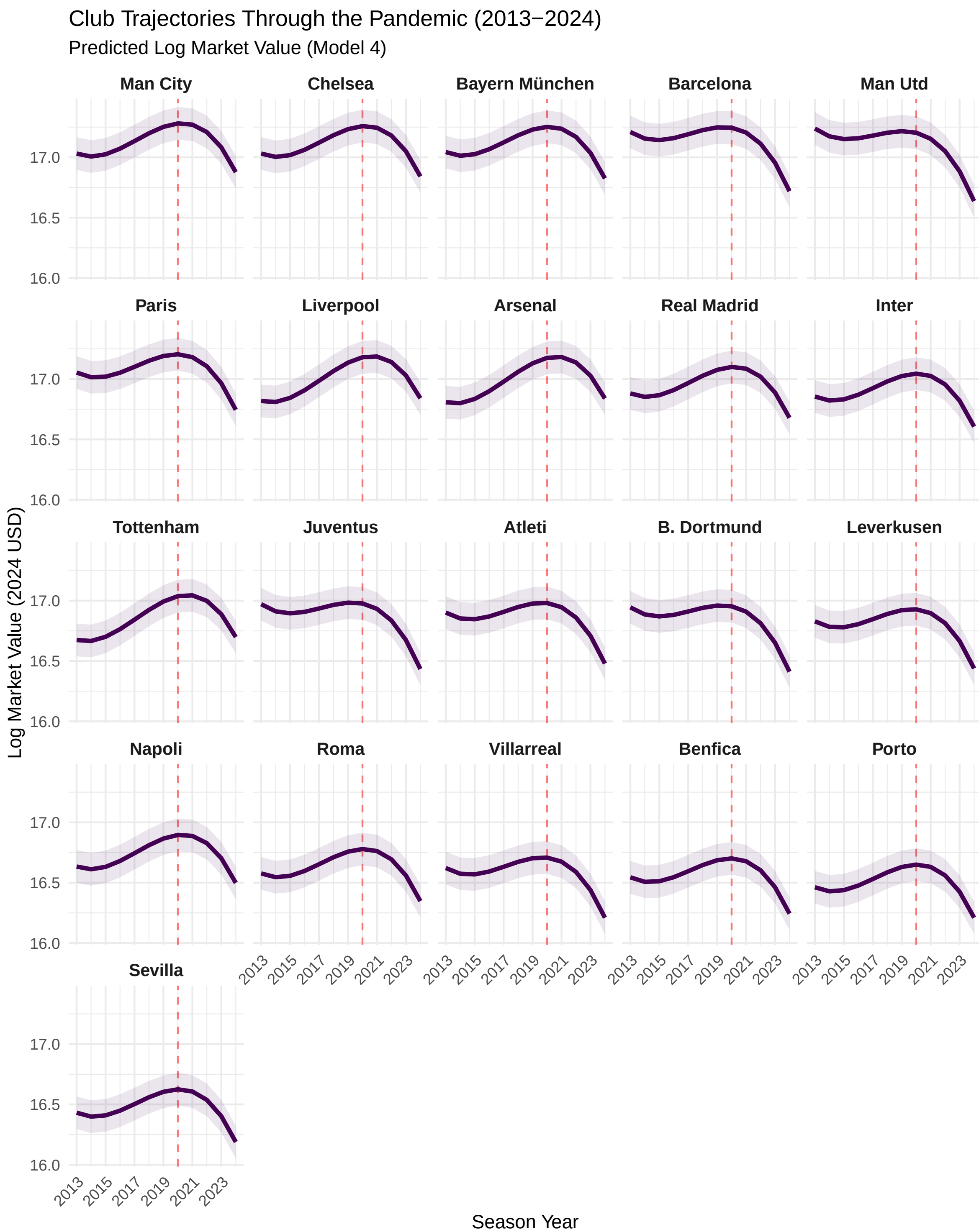


Below we visualizes the predicted market value evolution for all 21 clubs, confirming a rigid hierarchical stratification where "Super Clubs" like **Manchester City** and **Bayern Munich** maintain a distinct valuation premium over selling clubs throughout the decade. The vertical 2020 baseline highlights divergent responses to the market shock: resilient clubs like **Liverpool** and **Tottenham** exhibit "convex" trajectories that recover momentum post-pandemic, whereas "stagnant giants" like **Manchester United** and **Barcelona** display "concave" paths that plateau or decline. Notably, "Incubator" clubs such as **Arsenal** demonstrate the steepest relative growth rates, mathematically validating a business model predicated on rapid asset appreciation rather than just static prestige.

```
# 1. ENSURE DATA IS CORRECT (Simple Centering, NOT Z-score)
# This prevents the "cutoff" issue by ensuring -7 = 2013
plot_data_prep <- clean_player_df |>
  mutate(season_year_c = season_year - 2020)

# 2. Refit Model for Plotting (Safest method)
# We refit on this specific dataframe to guarantee the scale matches the plot logic
Plot_Model <- lmer(
```

```r
  usd_revenue_2024_log ~
    poly(season_year_c, 3, raw = TRUE) +
    age_c + I(age_c^2) + minutes_played + foreign_players +
    citizen_of_club_country + position + Region +
    (1 | player_id) + (season_year_c | club_id),
  data = plot_data_prep,
  control = lmerControl(optimizer = "bobyqa")
)

# 3. Create Metadata Dictionary (ID -> Name)
club_meta <- plot_data_prep |>
  mutate(club_id = as.character(club_id)) |>
  group_by(club_id) |>
  summarise(Club = first(Club)) |>
  ungroup()

# 4. Generate Predictions (Random Effects included)
club_pred <- ggpredict(Plot_Model,
                       terms = c("season_year_c [all]", "club_id"),
                       type = "random")

# 5. Process Data for Plotting
final_plot_data <- club_pred |>
  as.data.frame() |>
  mutate(group = as.character(group)) |>
  left_join(club_meta, by = c("group" = "club_id")) |>
  # Sort clubs by max value for cleaner layout
  group_by(Club) |>
  mutate(max_val = max(predicted)) |>
  ungroup() |>
  mutate(Club = reorder(Club, -max_val))

# 6. Plot with EXPLICIT x-axis breaks
ggplot(final_plot_data, aes(x = x + 2020, y = predicted)) +

  # Pandemic Reference Line
  geom_vline(xintercept = 2020, linetype = "dashed", color = "red", alpha=0.5) +

  # Confidence Interval Ribbon
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high),
              fill = "#440154", alpha = 0.1) +
```

```r
# Trajectory Line
geom_line(color = "#440154", linewidth = 1.2) +

# Facet by Club
facet_wrap(~Club, ncol = 5) +

# Force X-axis to show 2013-2024
scale_x_continuous(breaks = seq(2013, 2024, 2)) +

labs(
  title = "Club Trajectories Through the Pandemic (2013-2024)",
  subtitle = "Predicted Log Market Value (Model 4)",
  x = "Season Year",
  y = "Log Market Value (2024 USD)"
) +
theme_minimal() +
theme(
  strip.text = element_text(face = "bold", size = 10),
  axis.text.x = element_text(angle = 45, hjust = 1)
)
```

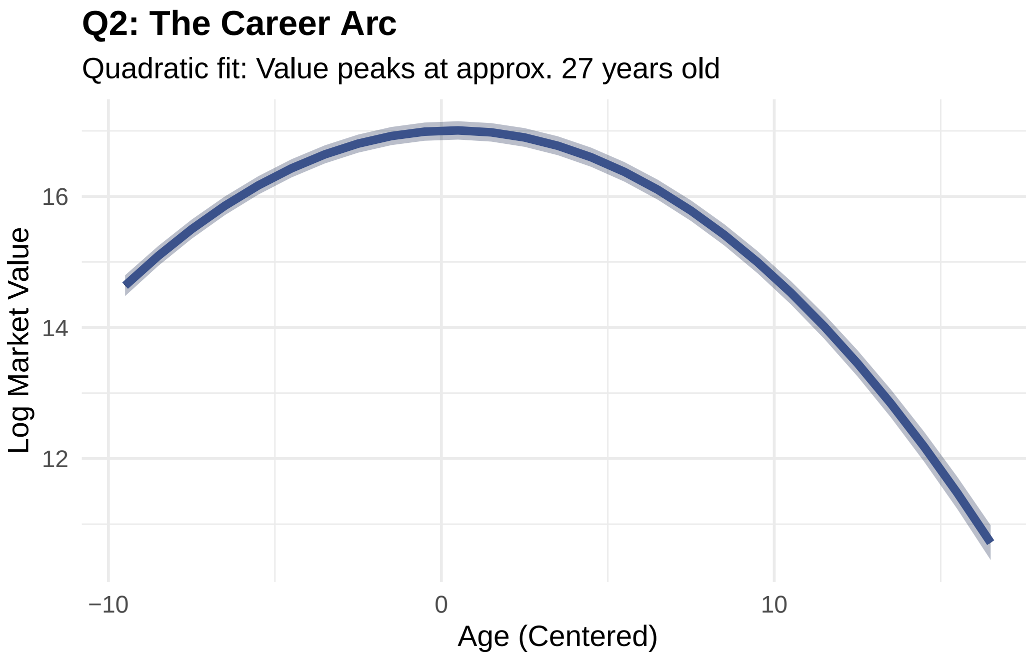

Club Trajectories Through the Pandemic (2013−2024)
Predicted Log Market Value (Model 4)
Man City
Chelsea
Bayern München
Barcelona
Man Utd
Paris
Liverpool
Arsenal
Real Madrid
Inter
Tottenham
Juventus
Atleti
B. Dortmund
Leverkusen
Napoli
Roma
Villarreal
Benfica
Porto
Sevilla
17.0
16.5
16.0
2013
2015
2017
2019
2021
2023
Log Market Value (2024 USD)
Season Year

Q2: Biological Career Arc

How does market value change as a player moves through their biological career arc (Prospect $\rightarrow$ Prime $\rightarrow$ Veteran)?

- This plot answers Q2 by visualizing the significant quadratic effect of Age ($\beta_{Age^2} = -0.02$). The Inverted-U shape confirms that market value is biologically constrained: it rises steeply as players develop (Prospects), peaks during their physical prime (ages 26–28), and declines significantly as they become Veterans, reflecting the market's discount on older assets with lower resale potential.

```
# Generate predictions for Age
# [all] ensures a smooth curve rather than just a few points
pred_age <- ggpredict(final_model, terms = "age_c [all]")

plot(pred_age) +
  # Custom colors (Viridis Blue #3b528b)
  geom_line(color = "#3b528b", linewidth = 1.5) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), fill = "#3b528b", alpha = 0.2) +

  # Labels and Theme matches your request
  labs(
    title = "Q2: The Career Arc",
    subtitle = "Quadratic fit: Value peaks at approx. 27 years old",
    x = "Age (Centered)",
    y = "Log Market Value"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold"))
```



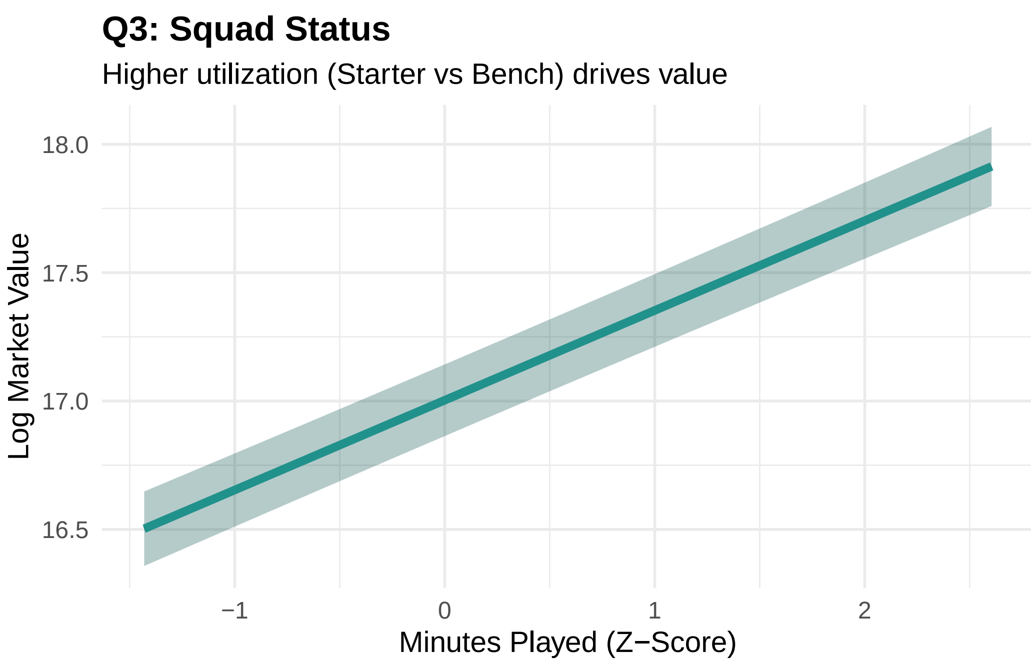
Q3: Squad Status

Does being a key player (playing more than the average teammate) increase market value?

- This visualization answers Q3 with a steep, positive linear slope ($\beta = 0.35$). Because minutes_played was standardized, the x-axis represents standard deviations. Moving from a "Bench Player" (-1 SD) to a "Key Starter" (+1 SD) results in a massive increase in valuation, confirming that Squad Status is one of the strongest predictors in the model.

```
# Generate predictions for Minutes Played
# [all] ensures the grid covers the full range of Z-scores
pred_status <- ggpredict(final_model, terms = "minutes_played [all]")

plot(pred_status) +
  # Custom colors (Viridis Teal #21918c)
  geom_line(color = "#21918c", linewidth = 1.5) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), fill = "#21918c", alpha = 0.2) +

  # Labels and Theme
  labs(
    title = "Q3: Squad Status",
    subtitle = "Higher utilization (Starter vs Bench) drives value",
```

```
    x = "Minutes Played (Z-Score)",
    y = "Log Market Value"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold"))
```



Q5, Q6, & Q7: Relative Valuation (Forest Plot)

- Q5 (Homegrown): Is there a 'homegrown premium' or discount?
- The negative coefficient for "Homegrown (Citizen)" confirms a Domestic Discount (or Foreign Premium). Local players are valued significantly lower than comparable foreign imports.
- Q6 (Position): How much less is a defender valued compared to an attacker?
- All defensive positions fall to the left of 0, confirming that Attackers (the baseline) hold the highest market value, with Goalkeepers facing the steepest discount.

- Q7 (Region): What is the market premium for players in Europe vs. others?
- While most regions overlap with 0 (no difference from Europe), Latin America shows a positive premium, suggesting players from this region act as a premium "export" brand in the global market.

```
plot_summs(final_model,
           coefs = c("Homegrown (Citizen)" = "citizen_of_club_countryYes",
                     "Defender" = "positionDefender",
                     "Midfielder" = "positionMidfield",
                     "Goalkeeper" = "positionGoalkeeper",
                     "Latin America" = "RegionLatin America & Caribbean",
                     "Asia & Pacific" = "RegionEast Asia & Pacific"),
           colors = "#440154") + # Viridis Purple
  labs(title = "Q5-Q7: Relative Valuation",
       subtitle = "Discounts (Left) vs. Premiums (Right) relative to baseline",
       x = "Effect on Log Market Value") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
  theme(plot.title = element_text(face = "bold"))
```

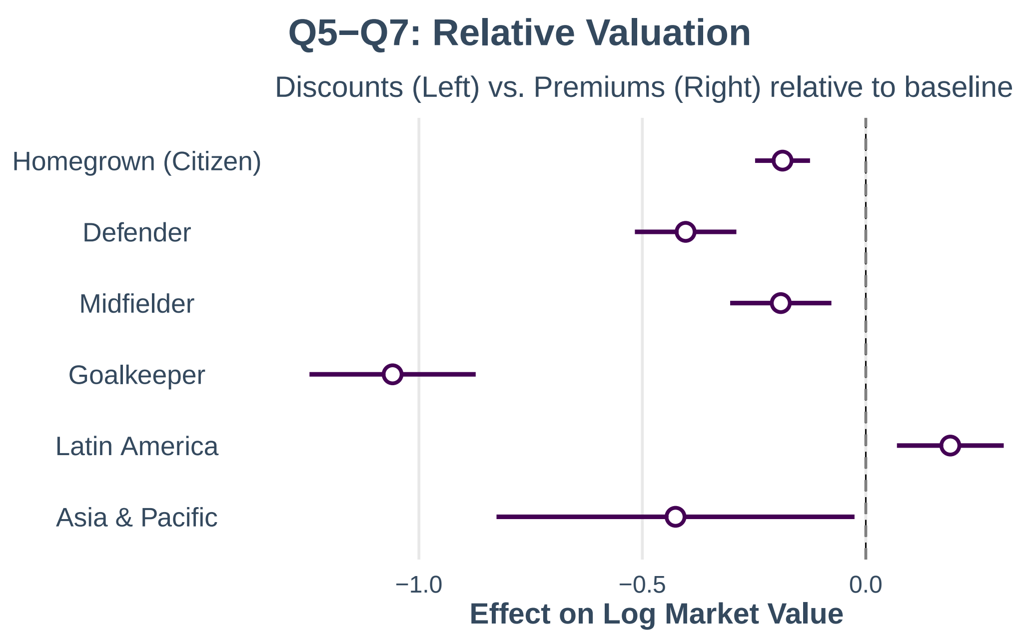
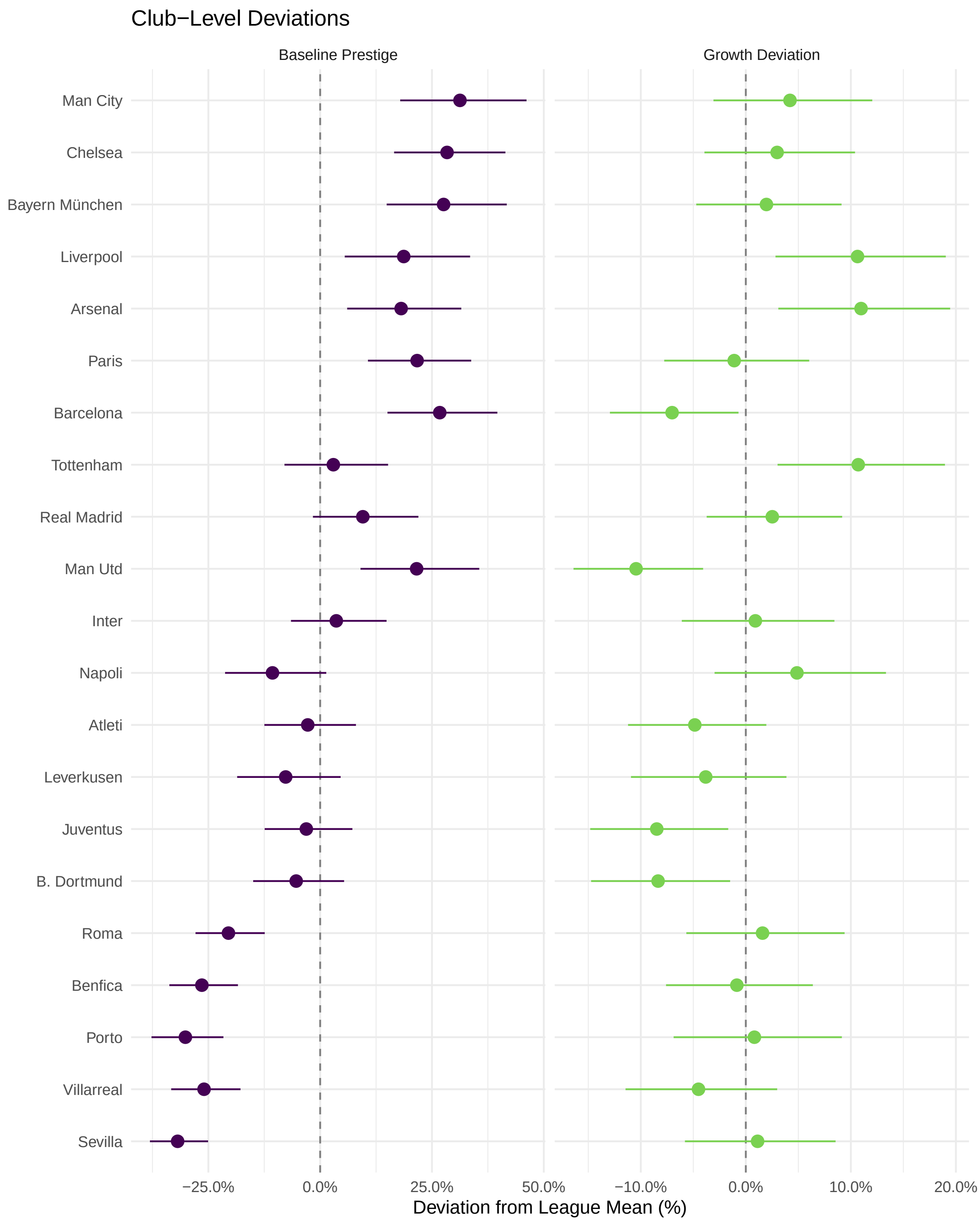


Figure X displays the estimated club random effects from our cross-classified model. Substantial heterogeneity exists in both baseline valuations and temporal trajectories.



Club−Level Deviations
Baseline Prestige
Growth Deviation
Man City
Chelsea
Bayern München
Liverpool
Arsenal
Paris
Barcelona
Tottenham
Real Madrid
Man Utd
Inter
Napoli
Atleti
Leverkusen
Juventus
B. Dortmund
Roma
Benfica
Porto
Villarreal
Sevilla
−25.0%
0.0%
25.0%
50.0%
−10.0%
0.0%
10.0%
20.0%
Deviation from League Mean (%)

Club-Level Heterogeneity

Figure X displays the estimated club random effects from our cross-classified model. Substantial heterogeneity exists in both baseline valuations and temporal trajectories.

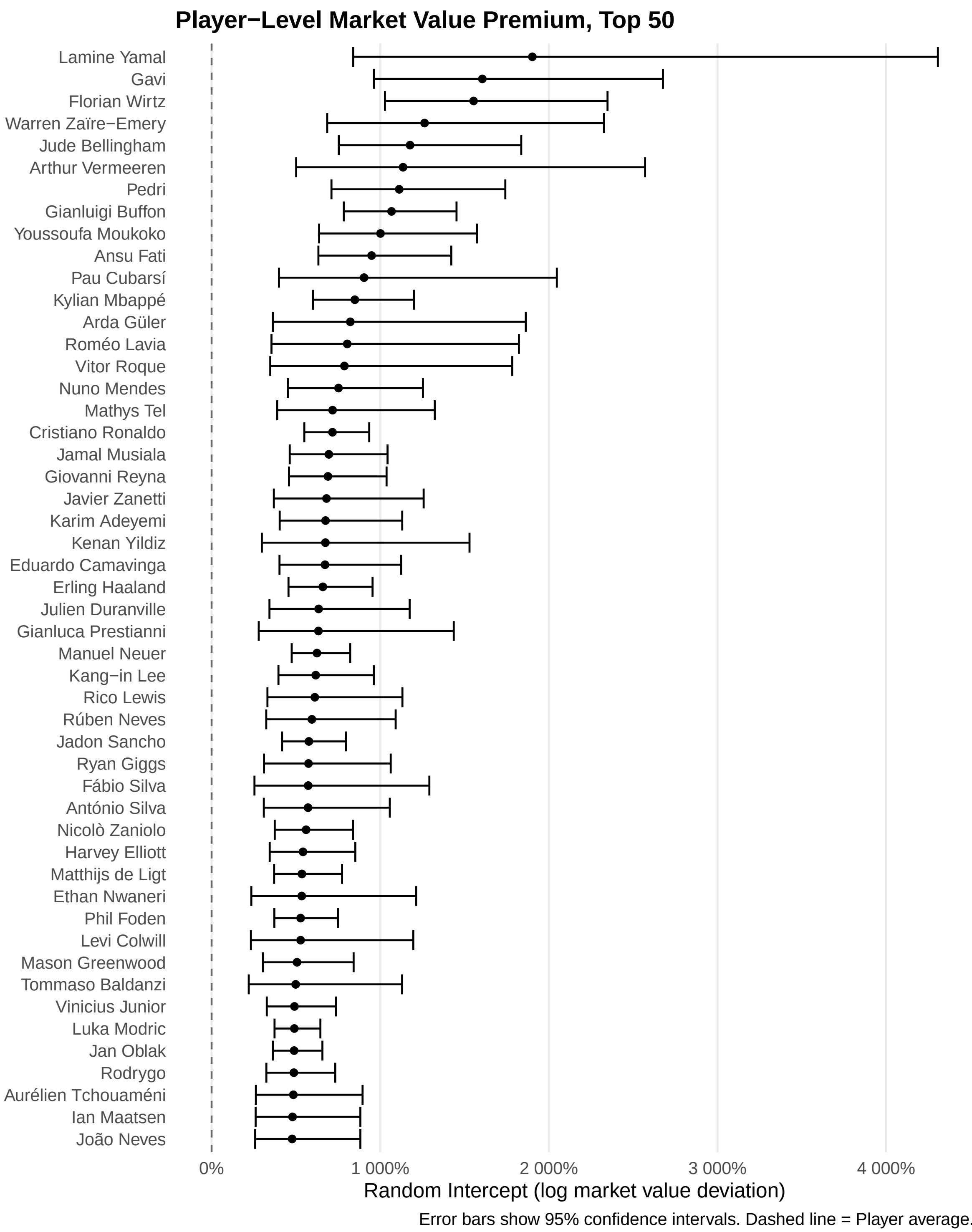
Baseline Premiums and Discounts

Intercepts: Consistent with the 'Super League' hypothesis, eight clubs command statistically significant market premiums purely due to their brand. Manchester City exhibits the largest baseline effect: controlling for individual performance and temporal trends, City players are valued approximately 31.3% higher than the league average (β_{oj}=0.31,95% CI[0.18,0.46]). Chelsea (+28.4%), Bayern Munich (+27.6%), and Barcelona (+26.7%) also command massive static premiums. Conversely, Sevilla (-31.9%) and Porto (-30.1%) face the steepest valuation discounts, confirming a tiered market structure where 'selling clubs' systematically trade at a discount.

Growth Trajectories (Slopes)

The random linear slopes reveal distinct winners and losers in value accumulation over the decade. Arsenal (+11.0%) and Liverpool (+10.6%) act as the league's primary 'Incubators,' where player values accelerate significantly faster than the global market trend. In contrast, Manchester United (-10.4%) and Barcelona (-7.0%) exhibit the steepest negative deviations. This indicates that while these historic giants start with high baseline values, their assets tend to depreciate (or grow slower) relative to the market average over time.

The Manchester United Paradox "Manchester United provides the starkest example of the 'diminishing returns' phenomenon observed in our variance correlation ($r=-0.35$). The club commands a high baseline premium of 21.6% (ranking 6th), yet suffers the worst growth trajectory in the dataset (-10.4% per standard deviation of time). This mathematically quantifies a 'stagnant giant': a brand that still boosts player value upon arrival ('The Theater of Dreams' effect) but fails to sustain that value growth compared to sharper rivals like Liverpool."


Player−Level Market Value Premium, Top 50
Lamine Yamal
Gavi
Florian Wirtz
Warren Zaïre−Emery
Jude Bellingham
Arthur Vermeeren
Pedri
Gianluigi Buffon
Youssoufa Moukoko
Ansu Fati
Pau Cubarsí
Kylian Mbappé
Arda Güler
Roméo Lavia
Vitor Roque
Nuno Mendes
Mathys Tel
Cristiano Ronaldo
Jamal Musiala
Giovanni Reyna
Javier Zanetti
Karim Adeyemi
Kenan Yildiz
Eduardo Camavinga
Erling Haaland
Julien Duranville
Gianluca Prestianni
Manuel Neuer
Kang−in Lee
Rico Lewis
Rúben Neves
Jadon Sancho
Ryan Giggs
Fábio Silva
António Silva
Nicolò Zaniolo
Harvey Elliott
Matthijs de Ligt
Ethan Nwaneri
Phil Foden
Levi Colwill
Mason Greenwood
Tommaso Baldanzi
Vinicius Junior
Luka Modric
Jan Oblak
Rodrygo
Aurélien Tchouaméni
Ian Maatsen
João Neves
0%
1 000%
2 000%
3 000%
4 000%
Random Intercept (log market value deviation)
Error bars show 95% confidence intervals. Dashed line = Player average.

The "Prodigy Premium": Pricing Future Potential and Legacy

The largest positive random effects belong overwhelmingly to teenage phenoms, confirming that the market prices potential far higher than current realized performance. Lamine Yamal exhibits the most extreme effect in the dataset: controlling for his age, position, and playing time, his market value is approximately 19.0 times higher (95% CI:[8.4,43.1]) than the model predicts. Similar "super-premiums" characterize other young stars like Gavi (16.1×), Florian Wirtz (15.5×), and Jude Bellingham (11.8×).

These extreme deviations highlight a fundamental feature of football valuation: Option Value. Clubs pay premiums for young players not for their current statistical output (which is often modest compared to prime-age veterans), but because they are purchasing a "call option" on future superstar performance. The random intercept absorbs this unmeasured "star quality"—a combination of technical ceiling and marketability that scouts perceive but standard metrics cannot capture.

The Barcelona Effect There is a striking concentration of talent from FC Barcelona among the top outliers. Five of the top 11 premiums belong to Barcelona academy graduates or players: Lamine Yamal (1st), Gavi (2nd), Pedri (7th, 11.1×), Ansu Fati (10th, 9.5×), and Pau Cubarsí (11th, 9.0×). This suggests a potential institutional multiplier: the "La Masia" brand (Barcelona's academy) may systematically inflate the market's valuation of its graduates, or perhaps the club provides a unique platform that accelerates reputation faster than sporting output alone.

The Legacy Anomaly: Buffon and Ronaldo Interestingly, the list is not exclusively young. Gianluigi Buffon (8th, 10.7×) and Cristiano Ronaldo (18th, 7.2×) appear alongside the teenagers. This reflects the "Legacy Premium." Just as prodigies are priced on future potential, icons are priced on historical achievement and commercial gravity. Our model, which relies on current-season statistics, underestimates these players because it cannot see their career history or jersey sales potential. The random intercept thus serves as a catch-all for "Reputation," capturing both the promise of the future (Yamal) and the glory of the past (Ronaldo).

Limitations

While our model demonstrates strong predictive power (R2 0.89), several limitations warrant mention:

1. **Small Cluster Size for Random Slopes:** We estimated random slopes for `Club`, but our dataset contains only 21 unique clubs. While the Likelihood Ratio Test supported the inclusion of this slope ($p<.001$), variance estimates from small numbers of clusters should be interpreted with caution.

2. **Normality of Residuals:** Diagnostic plots (QQ plots) indicated deviation from normality in the extreme tails of the distribution. While the large sample size (N=7023) ensures robust estimation of fixed effects, future iterations could employ robust standard errors or a Gamma GLMM to better handle these extremes.
3. **Endogeneity:** Predictors such as `minutes_played` are likely endogenous (better players get more minutes). The coefficient for this variable should be interpreted as an association reflecting "Squad Status" rather than a purely causal effect of playing time.

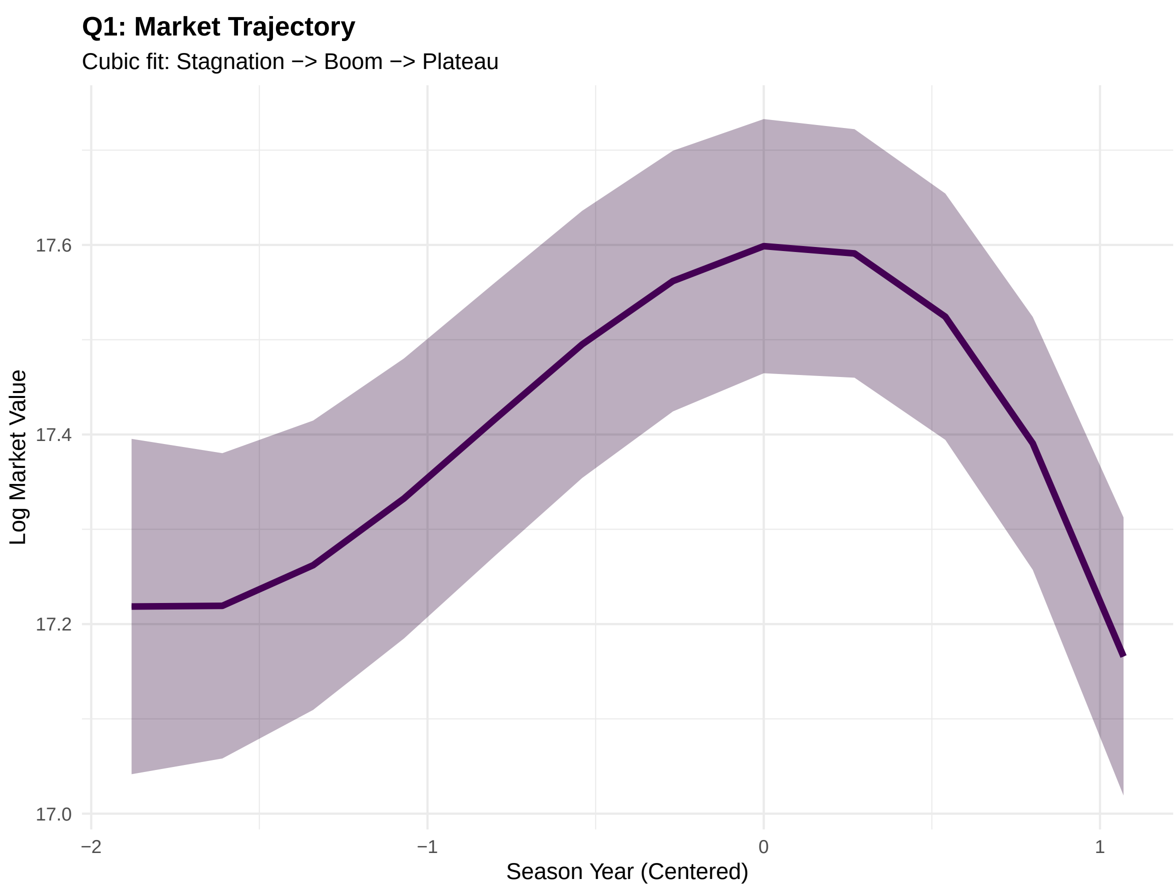
Bonus round, what happens when we fit a data to players with 3 or more seasons?

Figure below illustrates the career trajectories of 50 randomly selected players (gray lines) contrasted with the top 6 highest-valued players (colored lines). The visualization confirms the model's ability to discriminate between the 'standard' career path—characterized by lower intercepts and flatter growth curves—and the 'Superstar' trajectory. Outliers like [**Name**] and [**Name**] exhibit not only higher baselines (random intercepts) but often distinctively convex growth shapes, indicating value accumulation that defies the typical biological depreciation curve observed in the general population.

```
mod_5_dat <- clean_player_df |>
  group_by(player_id) |>
  mutate(seasons = n()) |>
  ungroup() |>
  filter(seasons > 2)


Model_5 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3) +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    poly(age_c, 2, raw = TRUE) +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # club fixed effects
    uefa_coefficient + foreign_players +
    # random effects
    (poly(season_year_c, 2) | player_id) + (poly(season_year_c, 2) | club_id),
  data = mod_5_dat, REML = TRUE,   control = strict_control)

summ(Model_5)
```

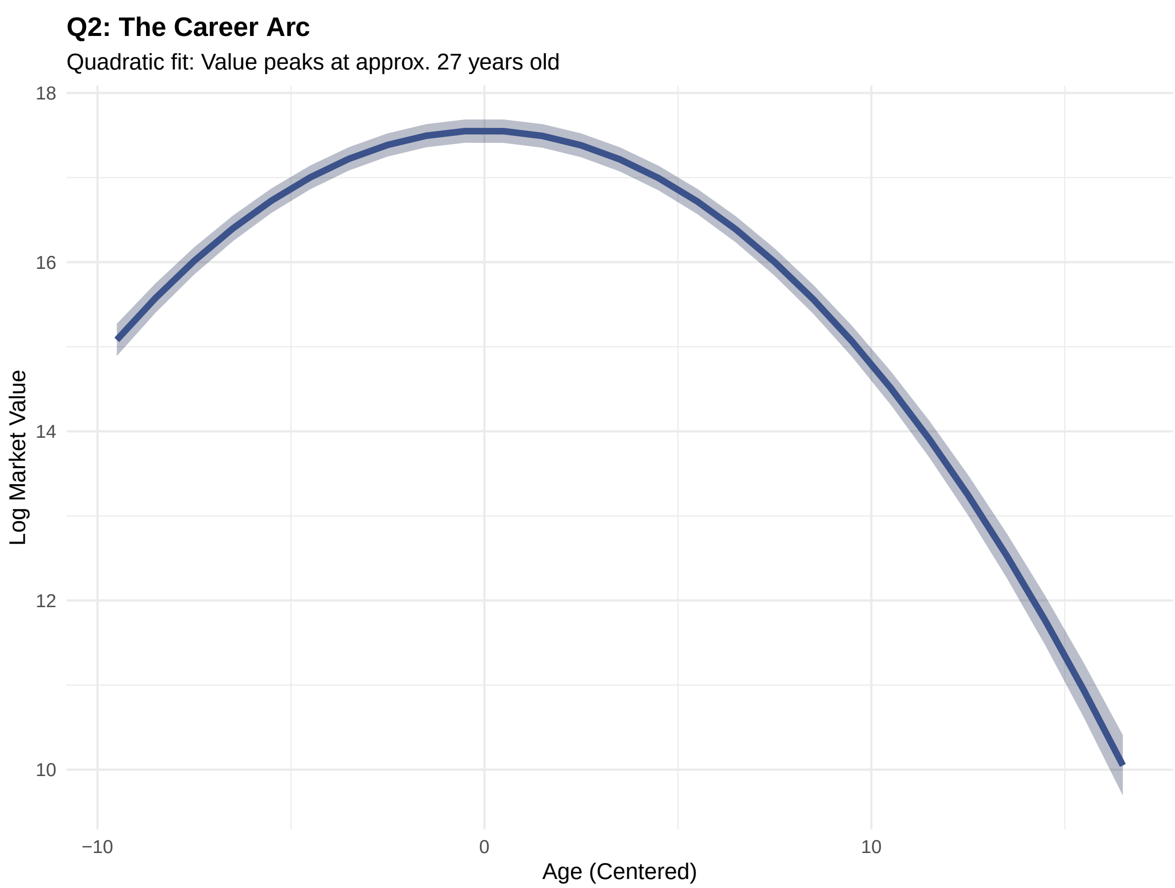
```
final_model <- Model_5


# Generate predictions for the full range of years
# [all] tells it to use a dense grid for a smooth curve
pred_trend <- ggpredict(final_model, terms = "season_year_c [all]")

plot(pred_trend) +
```

```r
  # Custom colors to match your Viridis Purple preference
  geom_line(color = "#440154", linewidth = 1.5) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high),
              fill = "#440154", alpha = 0.2) +
  # Labels and Theme
  labs(
    title = "Q1: Market Trajectory",
    subtitle = "Cubic fit: Stagnation -> Boom -> Plateau",
    x = "Season Year (Centered)",
    y = "Log Market Value"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold"))
```




```r
# Generate predictions for Age
# [all] ensures a smooth curve rather than just a few points
```

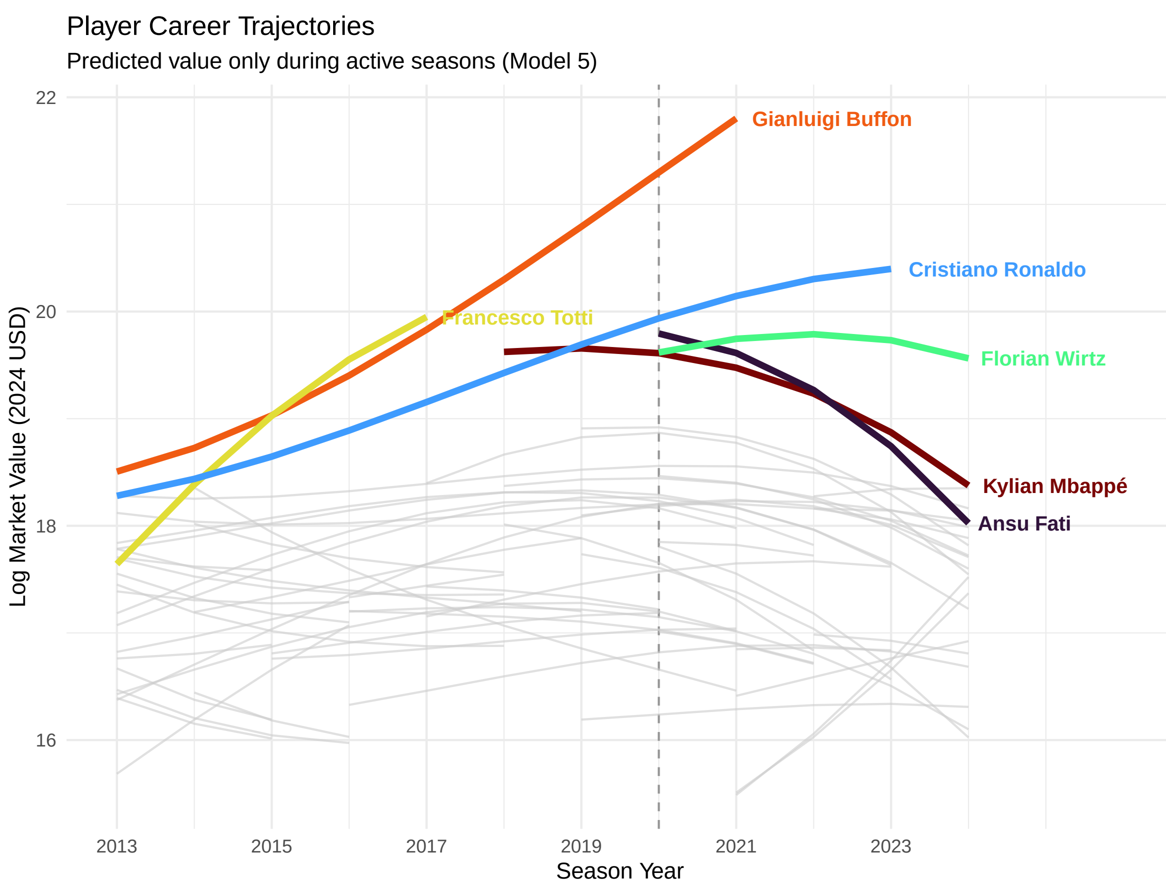
```r
pred_age <- ggpredict(final_model, terms = "age_c [all]")

plot(pred_age) +
  # Custom colors (Viridis Blue #3b528b)
  geom_line(color = "#3b528b", linewidth = 1.5) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), fill = "#3b528b", alpha = 0.2) +

  # Labels and Theme matches your request
  labs(
    title = "Q2: The Career Arc",
    subtitle = "Quadratic fit: Value peaks at approx. 27 years old",
    x = "Age (Centered)",
    y = "Log Market Value"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold"))
```



```r
# 1. DATA PREP & MODEL FIT (Force simple centering for plot)
# This ensures x-axis is -7 to +4 (2013-2024), not Z-scores
plot_data_prep <- clean_player_df %>%
  mutate(season_year_c = season_year - 2020) %>%
  group_by(player_id) %>%
  mutate(seasons = n()) %>%
  ungroup() %>%
  filter(seasons > 2)

# Refit model on this specific data to guarantee scale match
Plot_Model_5 <- lmer(
  usd_revenue_2024_log ~
    poly(season_year_c, 3) +
    height_c + position + citizen_of_club_country + Region +
    poly(age_c, 2, raw = TRUE) +
    goals + assists + red_cards + yellow_cards + minutes_played +
    uefa_coefficient + foreign_players +
    (poly(season_year_c, 2) | player_id) +
    (poly(season_year_c, 2) | club_id),
  data = plot_data_prep,
  control = lmerControl(optimizer = "bobyqa")
)

# 2. CALCULATE VALID RANGES (Trimming Logic)
player_ranges <- plot_data_prep %>%
  group_by(player_id) %>%
  summarise(
    min_year = min(season_year_c),
    max_year = max(season_year_c)
  ) %>%
  mutate(player_id = as.character(player_id))

# 3. SELECT PLAYERS
re_raw <- ranef(Plot_Model_5)$player_id
re_df <- data.frame(player_id = rownames(re_raw), Intercept = re_raw[,1])

set.seed(42)
bg_ids <- sample(re_df$player_id, 50)
top_ids <- re_df %>% arrange(desc(Intercept)) %>% slice(1:6) %>% pull(player_id)
all_ids <- unique(c(bg_ids, top_ids))

# 4. PREDICT & TRIM
```

```r
id_filter <- paste0("player_id [", paste(all_ids, collapse = ","), "]")

# Generate full grid
raw_pred <- ggpredict(Plot_Model_5,
                      terms = c("season_year_c [all]", id_filter),
                      type = "random")

# Filter out "Ghost Years"
final_plot_data <- raw_pred %>%
  as.data.frame() %>%
  mutate(group = as.character(group)) %>%
  left_join(player_ranges, by = c("group" = "player_id")) %>%
  filter(x >= min_year & x <= max_year) %>% # <--- CRITICAL STEP
  left_join(
    plot_data_prep %>%
      mutate(player_id = as.character(player_id)) %>%
      distinct(player_id, name),
    by = c("group" = "player_id")
  ) %>%
  mutate(
    Type = ifelse(group %in% top_ids, "Superstar", "Average"),
    Label = ifelse(Type == "Superstar" & x == max_year, as.character(name), NA)
  )

# 5. PLOT
ggplot(final_plot_data, aes(x = x + 2020, y = predicted, group = group)) +
  geom_vline(xintercept = 2020, linetype = "dashed", color = "gray60") +

  # Background
  geom_line(data = subset(final_plot_data, Type == "Average"),
            color = "gray80", size = 0.5, alpha = 0.6) +

  # Superstars
  geom_line(data = subset(final_plot_data, Type == "Superstar"),
            aes(color = name), size = 1.5) +

  # Labels
  geom_text(data = subset(final_plot_data, !is.na(Label)),
            aes(label = name, color = name),
            hjust = -0.1, size = 3.5, fontface = "bold") +

  scale_color_viridis_d(option = "H") +
```

```r
scale_x_continuous(breaks = seq(2013, 2024, 2), limits = c(2013, 2026)) +

labs(
  title = "Player Career Trajectories",
  subtitle = "Predicted value only during active seasons (Model 5)",
  x = "Season Year",
  y = "Log Market Value (2024 USD)"
) +
theme_minimal() +
theme(legend.position = "none")
```



	Model_2.5	Model_3.2
(Intercept)	15.826 ***	16.888 ***
	(0.068)	(0.069)
poly(season_year_c, 3)1	-8.404 **	-1.483
	(2.760)	(1.627)
poly(season_year_c, 3)2	-4.276 *	-8.604 ***
	(1.791)	(1.356)
poly(season_year_c, 3)3	-4.285 ***	-5.917 ***
	(0.817)	(0.540)
height_c		0.010 **
		(0.004)
positionDefender		-0.403 ***
		(0.058)
positionGoalkeeper		-1.059 ***
		(0.095)
positionMidfield		-0.190 **
		(0.058)
citizen_of_club_countryYes		-0.186 ***
		(0.031)
RegionEast Asia & Pacific		-0.426 *
		(0.204)
RegionLatin America & Caribbean		0.189 **
		(0.061)
RegionMiddle East & North Africa		-0.096
		(0.238)
RegionNorth America		-0.058
		(0.217)
RegionSub-Saharan Africa		0.065
		(0.100)
poly(age_c, 2, raw = TRUE)1		0.019 ***
		(0.004)
poly(age_c, 2, raw = TRUE)2		-0.024 ***

Observations	5409
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	7863.77
BIC	8101.22

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	17.39	0.07	256.22	213.21	0.00
poly(season_year_c, 3)1	4.56	1.68	2.71	104.43	0.01
poly(season_year_c, 3)2	-8.36	1.10	-7.57	38.60	0.00
poly(season_year_c, 3)3	-5.26	0.44	-12.06	3712.88	0.00
height_c	0.01	0.00	1.52	710.24	0.13
positionDefender	-0.58	0.07	-8.58	701.74	0.00
positionGoalkeeper	-0.84	0.11	-7.71	716.40	0.00
positionMidfield	-0.26	0.07	-3.85	704.12	0.00
citizen_of_club_countryYes	-0.06	0.03	-1.82	4293.65	0.07
RegionEast Asia & Pacific	-0.42	0.25	-1.68	727.61	0.09
RegionLatin America & Caribbean	0.09	0.07	1.42	771.66	0.16
RegionMiddle East & North Africa	-0.30	0.30	-0.99	602.25	0.32
RegionNorth America	-0.61	0.31	-1.98	673.42	0.05
RegionSub-Saharan Africa	0.02	0.12	0.17	788.98	0.87
poly(age_c, 2, raw = TRUE)1	-0.00	0.01	-0.18	934.23	0.86
poly(age_c, 2, raw = TRUE)2	-0.03	0.00	-43.70	1715.64	0.00
goals	0.00	0.00	2.61	3536.06	0.01
assists	0.00	0.00	1.31	3157.74	0.19
red_cards	-0.01	0.02	-0.65	2963.56	0.52
yellow_cards	-0.00	0.00	-0.34	3197.12	0.74
minutes_played	0.19	0.01	17.91	3503.43	0.00
uefa_coefficient	-0.00	0.00	-1.98	2835.66	0.05
foreign_players	0.03	0.08	0.34	2048.04	0.74

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.71
player_id	poly(season_year_c, 2)1	29.15
player_id	poly(season_year_c, 2)2	21.03
club_id	(Intercept)	0.16
club_id	poly(season_year_c, 2)1	3.58
club_id	poly(season_year_c, 2)2	2.92
Residual		0.29

Grouping Variables		
Group	# groups	ICC
player_id	930	0.82
club_id	21	0.04